

SEARCHING, SEIZING and USING COMPUTER BASED EVIDENCE

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Acquisition of Evidence

1. What Crimes

The issues raised in this paper apply to any case where the evidence is contained within a computer. These cases run the gamut from thefts to drugs, from child pornography to homicide.

2. What Computers

One of the first questions that must be asked is whose computer are we talking about. These fall into three categories:

- [The Victim's Computer;
- [The Suspect's Computer;
- [The Computers of Third Parties.

It is important to distinguish between computers in this manner because the law distinguishes between them. For example:

- [In most circumstances a suspect has no expectation of privacy in evidence found in the victim's computer.¹
- [The Privacy Protection Act, which sets significant limits on searches of publishers (including computer-based publishers), does not usually apply when the alleged publisher is a target. 42 U.S.C. § 2000aa(a)(1)
- [The Electronic Communications Privacy Act, 18 U.S.C. § 2701 *et seq.* also sets limits on the access to stored electronic communication, but the limits only apply to records stored on third party service providers, not to records on the suspect's computer.
- [Searches of the suspect's computer will always be subject to the full scrutiny of the Fourth Amendment and Article I, §7 of the Washington Constitution.

3. Legal Limits on Searches

Legal limits on searches flow from three sources

- [Constitutional Limits (Federal and State);
- [Statutory Limits;
- [Limits Imposed by Court Rule.

The restrictions on searches have different impacts. Searches that violate constitutional protection are likely to be suppressed. Searches that violate court rules may be suppressed or some lesser sanction may be applied. Searches that violate statutory restrictions may be subject to suppression, but more typically, while providing a civil remedy to the aggrieved party, do not

¹See *State v. Townsend*, 147 Wn.2d 666 (2002) where the issue was whether the defendant's e-mail and chat messages he had sent to the victim/undercover officer, which were then stored on the undercover officer's computer were "private communications" under the Washington Privacy Act. RCW 9.73.030. The Court held they were private communications but that the very nature of e-mail and chat implies consent to their recording. The case focused on the Privacy Act, not the Fourth Amendment or Article I, §7 of the Washington Constitution.

provide suppression as a remedy.

Let's examine each of these limitations in turn.

A. Constitutional Limits

The Fourth Amendment guarantees protection against unreasonable searches and seizures.² Article I, §7 of the Washington Constitution provides similar but more extensive guarantees.³ The questions to be asked in the Fourth Amendment and Article I, §7 context are three:

- [Does the Fourth Amendment/Article I, §7 apply to the situation?
- [Is there a legal warrant?
- [If there is not a legal warrant does the search fall within the exceptions to the warrant requirement?

1) Does the Fourth Amendment/Article I, § 7 apply to the situation?

This involves two separate considerations:

- [Is there a reasonable expectation of privacy
- [Is government action involved?

a) Is there a reasonable expectation of privacy?

In determining whether there is a reasonable expectation of privacy in a place searched, courts look at two factors - the subjective expectation of the person asserting a privacy interest and whether society accepts that as objectively reasonable. *California v. Greenwood*, 486 U.S. 35, 39 (1988). Our courts have consistently ruled that the highest privacy interest is attached to private dwellings. *Welsh v. Wisconsin*, 466 U.S. 740, 748 (1984). A computer inside a private dwelling will likely be accorded a similar privacy status.

Expectation of privacy in the contents of a computer is less clear when the computer is not in the home and when the contents of the computer are routinely shared with members of the public.

Is there a diminished expectation of privacy in a laptop computer outside the home? Laptop computers are like automobiles by virtue of their mobility. Should they have the same diminished expectation of privacy as automobiles? Probably not. Laptops are similar to automobiles in their mobility, but dissimilar in the other two ways that courts have used to justify a lower standard for automobiles - subject to government regulation, and the public nature of automobile travel. *South Dakota v. Opperman*, 428 U.S. 364, 367-68 (1976).

²"The right of the people to be secure in their persons, houses, papers, and effects, against unreasonable searches and seizures, shall not be violated, and no Warrants shall issue, but upon probable cause, supported by Oath or affirmation, and particularly describing the place to be searched, and the persons or things to be seized."

³"No person shall be disturbed in his private affairs, or his home invaded, without authority of law." See *State v. Gunwall*, 106 Wn.2d 54, 720 P.2d 808 (1986) for a discussion of the broader protections provided by the Washington Constitution.

Is there a reasonable expectation of privacy by an employee in data stored on a work computer? The answer to that question is fact specific. This topic is discussed in more detail later in this paper in the section dealing with consent by third parties.

What about situations where the suspect's computer is used as a public bulletin board or an ftp site that allows public access? Is there a reasonable expectation of privacy in the areas that are open to the public? Highly unlikely. See *Maryland v. Macon*, 472 U.S. 463, 469 (1985) discussing the absence of a legitimate expectation of privacy in the books displayed to the public in a bookstore.

b) Is government action involved?

The exclusionary rule was not intended to and does not apply to the actions of private citizens or foreign law enforcement. *Burdeau v. McDowell*, 256 U.S. 465, 475, 41 S.Ct. 574, 576, 65 L.Ed. 1048, 1051 (1921), *Weeks v. United States*, 232 U.S. 383, 398, 34 S.Ct. 341, 346, 58 L.Ed. 652, 658 (1914).

i) Searches by non forum-state law enforcement

The silver platter doctrine arose in the early 1900's when federal search and seizure law was more restrictive than state search and seizure law, and suppression was a remedy for violation of the federal law. When state law enforcement officers conducted a search legal under state law but illegal under federal law, could the fruits of the search be used in federal courts? The answer was yes, if the federal authorities received the fruits on a silver platter, i.e., they were not so involved in the illegal search as to require suppression.

* * *

The essential principle underlying the development of this "silver platter" doctrine is that protections afforded by the constitution of a sovereign

entity control the actions only of the agents of that sovereign entity.

State v. Mollica, 554 A.2d 1315 (N.J. 1989)⁴

There is no reason why the same principal should not apply to computer searches.⁵

ii) Searches by private citizens

The *Mollica* court viewed a transfer from a federal official to a state official as analogous to a transfer from a private citizen to a state official. *Mollica* at 1325, 1328. Searches by private citizens, not acting as the agent of law enforcement, are not entitled to constitutional protection. This view is consistent with numerous federal and state cases.⁶ The results of searches of

⁴*Mollica* contains the New Jersey Supreme Court's superb discussion of the history of the silver platter doctrine. *Mollica* involved an FBI investigation of illegal bookmaking being conducted from rooms at Caesar's Palace in Atlantic City, New Jersey. Acting without a warrant the FBI obtained from Caesar's the telephone toll records for the suspect's room for the duration of his stay. The FBI subsequently turned this information over to the New Jersey State Police who used the information to obtain a warrant. Evidence used at trial was discovered during the search. The crucial issues before the New Jersey Supreme Court were whether the seizure of telephone records by the FBI without a warrant was illegal under state law, and if so, whether the products of the state search under a warrant based upon the telephone records should be suppressed.

After concluding that state law prohibited the seizure of toll records without a warrant, the court addressed the second issue. After discussing the history and rationale of the doctrine, court focused on the crucial issue of agency.

As with the earlier manifestations of the silver platter doctrine, and as seen in the numerous post-*Mapp* [*Mapp v. Ohio*, 367 U.S. 643, 81 S.Ct. 1684, 6 L.Ed.2d 1081] examples of interstate transfers of evidence, the salient factor continues to be agency *vel non* between the officers of the respective jurisdictions. The nature of the relationship between the officers participating in the search or seizure and the officers seeking to make use of such evidence is critical.

* * *

This case thus requires us to consider the implications of the silver platter doctrine and its key element: intergovernmental agency. An important aspect of this determination is whether for constitutional purposes the federal agents can be said to be acting under the "color of state law." The assessment of the agency issue necessarily requires an examination of the entire relationship between the two sets of government actors no matter how obvious or obscure, plain or subtle, brief or prolonged their interactions may be. The reasons and the motives for making any search must be examined as well as the actions taken by the respective officers and the process used to find, select, and seize the evidence.

Mollica, at 1328-29.

⁵In *State v. Paris*, 76 Wn. App. 1056 (Unreported Decision, 1994), a local detective accompanied the U.S. Marshal in the execution of a writ of seizure under the federal Copyright Act. The detective observed, but did not participate in the search. The appellate court reversed the trial court's determination that the detective's mere presence at the search scene made the Marshal his agent, and allowed the evidence under the silver platter doctrine.

⁶*United States v. Gomez*, 614 F.2d 643 (9th Cir. 1979), *United States v. Jennings*, 653 F.2d 107 (4th Cir. 1981), *United States v. Entringer*, 532 F.2d 634 (8th Cir. 1976), *United States v. Sullivan*, 544 F.Supp 701 (D. Maine, 1982), *State v. Blackshear*, 511 P.2d 1272 (Or. App. 1973), *State v. Cox*, 674 P.2d 1127 (N.M.App. 1983), *State v. Bembeneck*, 331 NW 2d 616 (Wis. App. 1983), *State v. Goodman*, 51 A.D.2d 1008, 380 N.Y.S. 2d 768 (1976), *State v. Smith*, 110 Wn.2d 658, 756 P.2d 722 (1988).

computers by private citizens (not acting as agents of law enforcement) should be admissible.⁷

2) Is there a legal warrant?

a) The application for the warrant must demonstrate probable cause to believe that evidence of a crime will be found in the place to be searched

i) Evidence of a crime

The application for the warrant and the warrant itself must state specifically the statute believed to have been violated. This requirement serves as a limit on the search. Courts have held that the failure to state the crime specifically in the warrant itself is fatal, even if the officer executing the warrant knew what crime was being investigated. *State v. Riley*, 121 Wn.2d 22, 846 P.2d 1365 (1993). See discussion of particularity requirement in subsection (c) below.

Appellate courts have made it clear that failing to establish a nexus between the crime and the items or location to be searched will result in suppression. In *State v. Nordlund*, 113 Wn. App. 171 (2002) the police were investigating several instances of attacks against young girls and sought to search the suspect's computer. In pertinent part the affidavit said: "[Nordlund] used a computer at this residence to access pornography and to communicate with others via E-mail. Therefore, the computer and any electronic storage media could likely be important evidence in this case regarding intent, dates and locations." The court noted there was no nexus between the alleged crimes and Nordlund's use of the computer to access pornography and send e-mails. "Rather, it appears that the State was fishing for some incriminating document which is precisely what the first and fourth amendment prohibit."

⁷ Remember, however, that the provisions of the Privacy Act (RCW 9.73) are more restrictive. That law prohibits both the State and individuals from illegal interceptions and recordings of private communications. If a private citizen obtains information in violation of the Privacy Act, that evidence is inadmissible and evidence obtained under a warrant where the warrant was based on information obtained by a private citizen in violation of the Privacy Act is also inadmissible. *State v. Faford*, 128 Wn.2d 476, 910 P.2d 427 (1996). See, however, *State v. O'Neill*, 103 Wn.2d 853, 700 P.2d 711 (1985) where the court disallowed the use of federal recordings (legal under federal statute but not under Washington's Privacy Act) as evidence, but allowed those recordings to form the basis for probable cause for issuance of a warrant. The *Faford* Court did not discuss *O'Neill*.

ii) Will be found in the place to be searched

The most commonly overlooked element in search warrants that have presented to me for review is this one. What is the probable cause for believing that evidence of the crime you are investigating will be found in the location you wish to search. This requires articulable facts pointing to the location, not some generic “where else would it be” explanation.

A unique issue arises in the context of computer searches. Assume that you have traced an offending email or computer attack back to a particular Internet Protocol (IP) address and through that to a specific Internet Service Provider (ISP). From the service provider you obtain account information about to whom that IP address was assigned at the time of the incident in question. The account identifies an individual, a physical address, a telephone number, and other information.

Clearly you have probable cause to believe that evidence of a crime will be found in the computer used to send the offending e-mail or other message. But do you have probable cause to believe that evidence will be found in the suspect’s house?

If the account with the ISP is a dial-up account the account holder can access the Internet via that account from any number of locations. If you are extremely lucky, and the ISP in question users caller ID, and the caller ID information for the call in which you are interested in is still on the ISP’s system, then you may have sufficient information to search the address associated with the phone number identified by the caller ID system.⁸ (If the caller is using a wireless connection to the Internet your chances of getting a valid warrant to search the address associated with the ISP, without significant additional information, are slim.)

Where the IP address is associated with a physical address (i.e., a networked computer, a DSL or cable modem connection) the physical relationship between the IP address and the physical location to be searched is more reliable.⁹

In one case I was involved in we had traced back a suspect to a local ISP. We wanted a warrant to search the suspect’s house. But with only the account information from the ISP (a dial-up account), we did not have probable cause to search the suspect’s house. The suspect had left copies of sent and read email on the server. We obtained copies of this email. In one message the suspect stated it was after 9:30 (in the morning) and he had to go to class.

The police staked out the suspect’s house and on several occasions saw him leave his house between 9:30 and 9:45 and drive to a local community college where he was a student.

When the email was coupled with the surveillance results I was satisfied that we now had probable cause to search the suspect’s house. We had evidence that the suspect used the email account in question from his home. This was sufficient to make this a valid search.¹⁰

⁸In *United States v. Grant*, 218 F.3d 72 (1st Cir. 2000), the court upheld a search of the defendant’s house based on an IP address that was assigned to the defendant’s account, but without specific proof that the account was being used from the address that was searched. The court upheld the search even in the face of evidence showing that one of the accesses to the Internet was to a phone number in Virginia, while the defendant lived in Maine.

⁹See *United States v. Hay*, 231 F.3d 630, (9th Cir. 2000), where the IP address in question was associated with the University of Washington. UW records showed the IP address was associated with a network port wired to a specific apartment.

¹⁰See *United States v. Kennedy*, 81 F.Supp.2d 1103 (D.Kan. 2000). After identifying the account associated with

In some circumstances you may be able to defend an already issued warrant where the affidavit established a reasonable presumption that the evidence would be found at the location that was searched.¹¹

b) Signed by a neutral and detached magistrate

The warrant must be signed by a magistrate who has authority to issue the warrant. Obviously the judge must occupy a position that has been properly created and has authority to issue warrants.¹² In addition, and more important for Internet crimes and investigations, the judge must have authority to authorize a search of the location covered by the warrant. In Washington, district court judges only have authority to authorize searches within their county. A Superior Court Judge's authority to authorize searches may be limited to the boundaries of Washington. If a Superior Court Judge's authority is so limited, a warrant issued by a Washington Judge to search an ISP in another state is beyond that judge's authority. If the defendant had a recognizable privacy interest in the evidence provided under the warrant, the evidence is subject to suppression.¹³

This is, however an undecided issue. It can be argued that such a warrant merely directs a Washington peace officer to execute the warrant and the officer executes it by serving it via fax.

The Eighth Circuit addressed this issue in a circuitous way in *United States v. Bach*, 310 F.3d 1063 (8th Cir. 2002).

The Court reversed a decision of the trial court suppressing evidence obtained from Yahoo by a search warrant issued by a state court in Minnesota. The warrant was faxed to Yahoo who complied by sending the requested information to the Minnesota authorities.

The trial court, noting that 18 U.S.C. §3105 requires that an authorized officer be present and acting in the warrant's execution when a third party assists in a search, concluded that the Minnesota officer was not present when Yahoo executed the search and thus the search was improperly executed under §3105. Although the government argued that §3105 does not apply to state officers executing state warrants when there is no federal involvement, the court held that

a particular IP address, the FBI agent called the suspect and asked him if he was satisfied with his Internet service. The suspect confirmed that he accessed the Internet at his home. Although this was a cable modem service with a physical relationship between the IP address and the physical address, this same technique could be used in a dial-up account case.

¹¹*State v. McReynolds*, 104 Wn. App. 560 (2001). Although the affidavit was found insufficient to justify the warrant in the case before them, Division III noted that there might be situations where an affidavit could establish probable cause by creating a reasonable inference that the evidence sought would be found in the location to be searched. The court referenced Wayne R. LaFare, Search and Seizure § 3.7(d) at 381-84 (2d ed. 1996) on this issue. LaFare noted that in the case of a burglary, theft or robbery in which valuable property was stolen, there can be a reasonable inference that the property will be found at the suspect's residence, vehicle or place of business, because such stolen property is neither inherently incriminating nor easily stashed outside a home or vehicle.

¹²In Washington a district court commissioner was found to lack the authority to issue search warrants when the office of district court commissioner was not properly created. *State v. Moore*, 73 Wn. App. 805, 871 P.2d 1086 (1994).

¹³California has a statute that requires ISPs located in California to accept subpoenas and warrants from out of state as if they were issued in California. California Penal Code 1524.2. While this requires the California ISP to respond to a non-California warrant, it does not make the warrant valid in the issuing state if the warrant is outside the issuing judge's authority. A few other states have similar laws.

the Fourth Amendment provided the same protection. The court stated: “While state officers executing a state warrant without any assistance from federal authorities may not be required to comply with section 3105, protections analogous to those provided for by section 3105 exist under the Fourth Amendment.”

The Eighth Circuit, in a footnote, recognized the increasingly common practice of serving search warrants for electronic data by fax, much like subpoenas. The Court noted that since the search warrant standard is more stringent than the subpoena standard, so long as the warrant passes muster under warrant requirements the fact that it is being served like a subpoena is of no significance. The Court, noting that the Fourth Amendment is governed by a “reasonableness” standard, held that official presence is simply one of many factors considered in determining the reasonableness of the execution of a search warrant. The Court also noted that civilian searches are sometimes more reasonable than searches by officer and, if conducted outside the presence of police, may increase the amount of privacy retained by the individual during a search.

The Court noted that no warrant was physically “served”, no persons or premises were searched in the traditional sense and there was no confrontation between Yahoo technicians and the suspect. Furthermore, the Court observed that the physical presence of an officer would not have aided the search (and might have hindered it), the technical expertise of yahoo’s technicians far outweighed that of the officers, the items “seized” were located on Yahoo’s property, there was a warrant signed by a judge authorizing the search and the officers complied with the requirements of the Electronic Communications Privacy Act. The Court concluded “All of these factors weigh in favor of the government and we therefore find that the search was constitutional under the Fourth Amendment’s reasonableness standard.”

c) Reasonably precise in describing the place to be searched and the items to be seized

The first thing to understand under this topic is that a reviewing court will look at the four squares of the warrant itself and not consider information provided in the affidavit in support of the warrant unless the affidavit is incorporated by reference in the warrant.¹⁴

Does your warrant need to specifically allow the search of a computer before you can search a computer at the search location? If officers are executing a warrant that does not specifically reference computers, may they search a computer if it could contain the item they are authorized to seize, without an additional warrant? Under existing law the answer is clear that they may search the computer without an additional warrant. See *United States v. Ross*, 456 U.S. 798, 102 S.Ct. 2157, 72 L.Ed.2d 572 (1982) (“A lawful search of fixed premises generally extends to the entire area in which the object of the search may be found and is not limited by the possibility that separate acts of entry or opening may be required to complete the search . . .” and *United States v. Hunter*, 13 F.Supp.2d 574 (D.Vt. 1998) (Upholding a search of computers

¹⁴Whether this means you should also leave a copy of the affidavit at the search site along with a copy of the warrant is an open question. *United States v. Towne*, 997 F.2d 537 (9th Cir. 1993) notes the different positions courts have taken on this issue, including: (1) incorporation by reference is by itself sufficient; (2) incorporation is unnecessary if the affidavit was available for the officers executing the warrant to refer to; (3) incorporation is necessary and the incorporated affidavit must accompany the warrant during execution; and (4) the affidavit must be incorporated in and physically attached to the warrant. *United States v. Hayes*, 794 F.2d 1348, 1355 (9th Cir. 1986), states the purpose of the accompanying affidavit as both to “limit the officer’s discretion and to inform the person subject to the search what items the officers executing the warrant can seize.” See also *State v. Riley*, 121 Wn. 2d 22, 846 P.2d 1365 (1993).

even though the warrant did not specify computers because the items covered by the warrant could be found in the computer. “A finding of probable cause is not predicated on the government’s knowing precisely how certain records are stored.”)

Nevertheless, the better practice is, if you anticipate searching a computer, you should mention that in your affidavit and you should be able to articulate with some detail what you are looking for.¹⁵ The more general your description and the broader your intrusion into the suspect’s private affairs, the more likely your evidence will be suppressed.

The particularity requirement in the computer search context has been addressed by Washington courts on several occasions. In *State v. Riley*, , 121 Wn. 2d 22, 28, 846 P.2d 1365 (1993) the Court noted that the degree of particularity may be achieved by specifying the suspected crime. Otherwise the warrant must contain some other means of limiting the items to be seized. In *State v. Askham*, 120 Wn. App. (3/30/04) the Court of Appeals held that even where a warrant does not specifically name the crimes under investigation, if the warrant (and affidavit if incorporate) describe the suspected criminal activity in detail, this description may serve to limit the scope of the warrant.

If you know computers will be present, you should include them as items to be searched. But what if you don’t know if computers will be present. We have already discussed that computers don’t have to be named for you to be able to search them (if the items you are authorized to seize could be found in the computer). Should you, as a routine matter, request permission to search any computers that may be at the search site?

There are two views on this. Including this language over time may create a *de facto* requirement that computers have to be listed or they can’t be searched. In an odd way, requesting permission to search computers on the site when you don’t have probable cause to believe that computers are there, may cause the search to be invalidated.¹⁶

Failure to include the language, however, may lead to suppression, particularly at the trial court level.

¹⁵For example here is an excerpt from a warrant I was involved with:

The computers and magnetic media seized may be searched for:

- [Files specific to the King County Library System, such as references to Dynix, the specific operating system used by the library, files containing library records, etc;
- [Files similar or identical to those stored by Phree on the library computer;
- [Messages and other substantive files referencing Phree, KoBK, the library, the method of access, etc;
- [Files showing Higgins dominion and control over the computer(s) being seized;
- [Files relating to Internet usage and familiarity;
- [Files relating to UNIX familiarity.

¹⁶Examine *United States v. Nafzger*, 965 F.2d 213 (7th Cir. 1992) where the warrant authorized the search of a specific stolen truck. The truck in question was in a tool shed. The police entered the shed and found the truck. The evidence was suppressed because there was “a total failure to show probable cause that the truck described could be found” at the location searched. Requesting a search of a computer when you don’t show probable cause to believe a computer will be found at the search site may be subject to the same remedy.

My suggestion is to include language like “including any computers and computer media located therein” in your warrant language describing where you are authorized to search (along with a definition of “records” or “documents” sufficient to cover electronic versions - see later discussion.) You still run the risk that judges will come to expect this, but by including a reference to computers and media in general rather than a reference to a specific computer you will diminish the likelihood of a *Nafziger* result.¹⁷

d) Knock and announce

In executing a warrant police are required to knock and announce their purpose. This is both a Fourth Amendment requirement (*Wilson v. Arkansas*, 514 U.S. 927, 115 S.Ct. 1914, 131 L.Ed.2d 976 (1995)), and a statutory requirement in Washington.¹⁸ The purpose of such a requirement is to reduce the potential for violence to all parties from unannounced entry, prevent unnecessary property damage, and protect the privacy rights of occupants.¹⁹

Computer data can be destroyed (or at least made more difficult to recover) in a short period of time. In the appropriate circumstances you may want to request authorization to execute the warrant without knocking and announcing. You must, of course, justify such a request by the facts of your particular case.

e) Take items in plain view

The plain view doctrine has two components. As an exception to the warrant requirement, contraband and instrumentalities may be seized without a warrant if they are in plain view and the incriminating nature of the item is immediately apparent. Related is the principle that while you are executing a warrant for one item you can seize other items in plain view.

This is discussed in more detail in the discussion of the plain view exception to warrant requirements below.

f) Good faith

The exclusionary rule does not apply in federal court when evidence is seized in reasonable, good faith, reliance on a search warrant that is later found to be unsupported by probable cause. The rationale behind this is that the deterrent function of the exclusionary rule is not furthered by penalizing an officer who has complied, in good faith with the warrant requirement. The exclusionary rule is, after all, intended to deter police misconduct, not to cure violations or rights.

The test is whether the officers acted in objectively reasonable reliance on what turned out to be an invalid warrant. *United States v. Leon*, 468 U.W. 897, 104 S.Ct. 34-5, 82 L.Ed.2d 677 (1984).

The good faith rule has been adopted by some, but not all courts. It has also been applied in non-criminal contexts.²⁰ It is NOT the rule in Washington.

¹⁷See n. 16 above.

¹⁸RCW 10.31.040.

¹⁹Note, 80 Yale L.J. 139, 140-42 (1970)

²⁰See the discussion of the Privacy Protection Act and the Electronic Communications Privacy Act below.

g) Take computers off premises to search

Often (and in these days of multi-gigabyte drives, usually) you will need to take a computer off-site to complete the search. So long as this has been justified and requested in the application and approved by the warrant judge, this is generally an accepted practice. As noted by District Court Judge Sessions in *U.S. v. Hunter*, 13 F.Supp.2d 574, 583 (D.Vt. 1998)

[U]ntil technology and law enforcement expertise render on-site computer records searching both possible and practical, wholesale seizures, if adequately safeguarded, must occur.”

A recent Colorado case²¹ followed the guidance of *Hunter* and other cases, but the dissent raised some issues you need to be prepared to deal with. In that case the police executed a warrant at the defendant’s apartment, searching for “written or printed material” relating to firearms, explosives, intent to do harm to other people, etc. They seized two desktop computers and five laptop computers and removed them from the premises for offsite searching. They then got a separate warrant to search the computers.

The trial court suppressed the evidence found on the computers because the court was unable to conclude that a computer was analogous to a writing or other type of document – the items that could be seized under the warrant. The Colorado Supreme Court reversed the trial court and ruled the evidence admissible, citing numerous cases found in this paper and in the Department of Justice’s Computer Search guidelines.²² They viewed the computers as containers which under well-established law,²³ could be searched if they could contain the evidence for which the warrant authorized seizure. The court specifically approved the procedure of seizing the computers for off-site search.

The dissent argued that computers were not like other containers and that special privacy interests were involved. In particular, the dissent argued that the case law did not support seizing computers to search off-site, even under a second warrant, if the computers were not identified in the first warrant. The dissent also argued that the subsequent search, off-site, should be designed to limit the privacy intrusion.

²¹*People v. Michael John Gall*, (Colo. Supreme, 3/5/2001).

²²DOJ Computer Search Guidelines: <http://www.cybercrime.gov/s&smanual2002.htm>

²³See the discussion of *United States v. Ross*, 456 U.S. 798, earlier in this paper.

h) Prompt execution

Finally, a warrant, to be valid, must be executed promptly. What is prompt execution? There are two components. First the warrant must be executed while there is still probable cause to believe that evidence of a crime will be found at the place to be searched - before the information provided in support of the warrant becomes stale. This is a constitutional issue. See LaFave, *Search and Seizure*, §4.7, pp. 584-588 (3d. Ed.)

A related but separate issue is when laws or court rules require a warrant to be executed within a certain amount of time, setting the outer limits on the time a magistrate may set for the execution of the warrant. This is discussed below in the section dealing with court rules and magistrate imposed restrictions

3) If there is not a legal warrant, does the search fall within the exceptions to the warrant requirement?

There are exceptions to the rule that evidence obtained without a warrant is inadmissible. These are, of course, the “few, specific, established, and well-delineated exceptions” developed under the Fourth Amendment. *Schneckloth v. Bustamonte*, 412 U.S. 218, 219, 36 L.Ed. 2d 834, 93 S. Ct. 2041 (1973), citing *Katz v. United States*, 389 U.S. 347, 357, 19 L.Ed.2d 576, 88 S. Ct. 507 (1967).

A few of these exceptions, like consent, may justify the search of a computer without a warrant, but most do not. The more pertinent exceptions are discussed below.

a) Consent

If the suspect or other person with authority over the property consents to a search, then a warrant is not required. The key questions are:

- [Who has the authority to consent?
- [Is the consent given voluntarily and intelligently?
- [Is the consent limited?

i) Who may consent?

Obviously the suspect may consent to a search of his own premises and effects, subject to such a consent being given voluntarily and intelligently (see below.) As for third party consent, the general rule is that one “who possesses common authority over or other sufficient relationship to the premises to effects sought to be inspected” has the authority to consent to a search. *United States v. Matlock*, 415 U.S. 164, 171 (1974).

In *United States v. Rith*, 164 F.3d 1323 (10th Cir. 1999) the Tenth Circuit stated the *Matlock* test in the disjunctive: A person may give effective consent if they have mutual use of the property by virtue of joint access OR if they have control of the property for most purposes of it.

The crucial analytical test is whether the defendant has a reasonable expectation of privacy in the premises or effects that another has consented to a search of. In most instances of shared premises and effects the courts have ruled there is a reduced expectation of privacy in the premises or things shared with another. *United States v. Ladell*, 127 F.3d 622, 624 (7th Cir.

1997). As the Ninth Circuit stated:

Although there is always the fond hope that a co-occupant will follow one's known wishes the risks remain. A defendant cannot expect sole exclusionary authority unless he lives alone, or at least has a special and private space within the joint residence."

United States v. Morning, 64 F.3d 531, 536 (9th Cir. 1995).

The consent given by a third party may not be valid if the area involved has been set aside, or rented, for the exclusive use of the defendant. *State v. Mathe*, 102 Wn.2d 537, 688 P.2d 859 (1984).

In the computer search context the issue of consent most commonly arises in searches where an employer, a parent, a spouse, or a co-user has consented to the search of a computer containing evidence against the suspect. In analyzing consent in the computer context it is important to distinguish valid consent to search a room from valid consent to search a container in that room, such as a computer. The *Matlock* factors must apply to the computer itself, not merely the room containing the computer before consent is effective.²⁴

(a) Employer

In the private employer setting, If the computer in question belongs to the employer and is provided to the employee for his/her use in work related activities, the employee's reasonable expectation of privacy is significantly reduced. This is particularly true if the employer has announced a policy that the employer may access such computers at any time.²⁵ In such a situation the employer may consent to a search of the employee's computer.

Similarly, if the computer in question is routinely used by multiple employees, any one of the users may consent to a search of the computer.

On the other hand, if the employer explicitly allows (or tolerates) use of a work computer for private purposes and the employee sets aside a separate area of the computer for her/his private files, the question is much closer. If the employee encrypts those files or otherwise restricts access to them, the expectation of privacy is higher still. The same is true (regarding segregated areas and encrypted files) for a shared computer.

In non-computer cases involving private employers, the courts have consistently held that the employer has the authority to consent to a search of those areas not set aside for exclusive use by a particular employee. *United States v. Gargiso*, 456 F.2d 583 (2d Cir. 1972).

But what about public employees? Do government workers have any reasonable expectation of privacy in their workplace?

²⁴"While authority to consent to search of a general area must obviously extend to most objects in plain view within the area, it cannot be thought automatically to extend to the interiors of every discrete enclosed space capable of search within the area." (While search of the room by consent was okay, search of a footlocker was not.) *United States v. Block*, 590 F.2d 535, 541 (4th Cir. 1978)

²⁵See *U.S. v. Simons*, 29 F.Supp.2d 324 (E.D. Va. 1998). The defendant was an employee at the CIA's Foreign Bureau of Information Services. The court held he did not have a reasonable expectation of privacy regarding his Internet use in light of his employer's written policy restricting Internet use.

In *O'Connor v. Ortega*, 480 U.S. 709, 94 L.Ed.2d 714, 107 S.Ct. 1492 (1987) the Supreme Court held that a government employee does not lose their expectation of privacy merely because they are public employees. The Court did note, however, that the nature of the public workplace can be such that the expectation of privacy is greatly reduced or eliminated in some circumstances. “Some government offices may be so open to fellow employees or the public that no expectation of privacy is reasonable.”

The Court went on to say, that even when there is a reasonable expectation of privacy, a government employer may search the files and offices of an employee, without consent or a warrant, for work-related reasons (to locate a particular document, for example) or to investigate suspected employee wrongdoing.²⁶

Two recent cases provide additional insight into the search of public employee computers. In *United States v. Slanina*, 283 F.3d 670 (5th Cir. 2002), the court held that a city employee’s expectation of privacy in his office computer was reasonable because the employer had not notified employees that their computer usage would be monitored and there was no indication that other employees had routine access to Slanina’s computer.

The court nevertheless upheld the search under the *O'Connor* exception permitting warrantless searches by an employer to investigate suspected employee malfeasance.

Specifically a member of the IT staff had accessed Slanina’s computer for network installation purposes and had uncovered evidence of pornography. The IT worker brought this to the attention of Slanina’s supervisor who happened to be a law enforcement officer with supervision over the fire department where Slanina worked. Child pornography was found on the computer and a criminal prosecution ensued.

The court upheld the search notwithstanding that the employee malfeasance in question also happened to be a crime and notwithstanding that the supervisor was also a law enforcement officer.

In the same week as this decision, the Tenth Circuit ruled that an Oklahoma State University Professor did not have a reasonable expectation of privacy in materials stored on his office computer. *United States v. Angevine* 281 F.3d 1130, 1134 (10th Cir. 2002). OSU had a computer use policy that explained appropriate computer use, warned employees about the consequence of misuse and described how officials administered and monitored the University computer network. This was done via a policy manual distributed to all employees as well as in a banner that was displayed each time the computer was turned on. This is a still developing area of law. See generally See LaFave, *Search and Seizure*, §8.6(f) (3d. Ed.)

(b) Parent

Courts generally find third-party consents easier to sustain “if the relationship between the parties - parent to child here, spouse to spouse in other cases - is especially close.” *United States v. Ladell*, 127 F.3d 622, 624 (7th Cir. 1997).

In the computer context, the more the computer is a shared item, both in terms of use and location, the more likely it is that a parent can give a valid consent. Even when the computer

²⁶For a computer specific case, where a supervisor’s seizure of a computer disk containing work related and personal files was upheld, see *Williams v. Philadelphia Housing Authority*, 826 F.Supp. 952 (E.D.Pa. 1993, *aff’d mem.* 27 F.3d 560 (3d Cir. 1994).

belongs to the child or is used by the child exclusively and is located in their room or private space, parents can generally consent to a search of such a computer so long as the child is “essentially dependent” on the parent. If the child pays rent or in other ways is independent of the parent, the parent may lack the authority to give valid consent to a search of the child’s effects or private space. *United States v. Rith*, *supra*.

(c) Spouse

Spouses are generally considered to have joint control and equal right to occupancy of the premises and access to computers on the premises, and so may give a valid consent.

If the computer is used exclusively by the non-consenting spouse and is kept in a separate room (especially if locked), the other spouse may not be able to give valid consent.

(d) Co-user

A co-user of a computer is much like a co-tenant and should be analyzed the same way. Generally, allowing co-use results in a greatly reduced expectation of privacy. Whether password restricted or encrypted areas are the equivalent to locked bedrooms and thus not subject to a consent search by a co-user is an as yet undecided issue.²⁷ Until you learn otherwise, however, I would assume that a co-user CANNOT give consent to search such objectively private areas of a computer.

(e) Other

When an item has been left in the custody of a third party the courts may find limits on the third party’s authority to consent to a search of the item. In *United States v. James*, ___ F.3d ___, 2003 WL 22998108 (8th Cir. Dec. 23, 2003) the Eighth Circuit held the third party did not have authority to consent to the search of a packet of computer CDs left in his possession by the defendant. The court focused on its perception of the defendant’s intent and held that the defendant did not intend to give the third party any authority to consent to the search of the disks. He gave them solely for the purpose of storing the disks.

This view of a suspect’s ability to limit the scope of authority given to a third party holding the defendant’s item conflicts with decisions like *United States v. Falcon*, 766 F.2d 1469 910th Cir. 1985). The defendant left a cassette at his brother’s apartment labeled “confidential” and “do not play”. Notwithstanding these instructions the *Falcon* court held the brother’s consent to search the cassettes to be effective because the brother “could have played the tape at any time, whether by design or by accident.”

One way of reconciling these two lines of thought is that in the *James* case the CDs were contained in a sealed envelope and the envelope was further encased in tape. Thus they could not have been viewed by accident. More importantly, the police came to third party in the *James* case because they intercepted a letter from James to the third party instructing the third party to

²⁷In a very fact specific ruling the court in *U.S. v. Smith*, 27 F.Supp.2d 1111 (C.C. Ill. 1998) allowed evidence obtained from a computer under a third party consent, noting that there was no evidence the defendant had password-protected the material seized.

The Fourth Circuit recently held that a search of an individual’s password-protected computer files is an illegal search in violation of the Fourth Amendment when consent to the search was given by another individual who had joint access to the computer but did not know the passwords and therefore did not have authority to grant access to the files. *Trulock v. Freeh*, 9 ILR (P&F) 3060 [4th Cir, December 28, 2001].

destroy the envelope and its contents.

Whether the specific facts of *James* will form the basis of future limits on consent searches is unknown, but the case does provide support for the more widely held view that where the defendant has taken extra steps to insure the privacy of computer files, such as by encrypting or password protecting them, the authority of the third party possessing or having access to the computer containing those files to consent to the search of those files is greatly limited. *Truloch v. Freeh*, 275 F.3d 391, 403 (4th Cir. 2001).

ii) Must the consensor be advised of his/her right to withhold consent?

The general rule is that the voluntariness of a consent to search is measured by the “totality of the circumstances.” *Schneckloth v. Bustamonte*, 412 U.S. 218, 367 L. Ed. 2d 854 (1968).

In the Federal system, the burden is on the government to prove voluntariness of consent by “preponderance of the evidence.” *U.S. v. Matlock*, 415 U.S. 164, 39 L. Ed. 2d 242 (1974), *U.S. v. Mendenhall*, 446 U.S. 544, 64 L. Ed. 2d 497 (1980). In Washington, a higher standard is required. Consent must be proved by “clear and convincing evidence” from the totality of the circumstances. *State v. Werth*, 18 Wn. App. 530, 571 P.2d 941 (1977).

Factors to be considered in determining the voluntariness of consent include:

- [The age of the suspect;
- [The level of intelligence of the suspect;
- [Whether the suspect was advised of her/his constitutional rights;
- [Whether the suspect was in custody, or detention, and the length;
- [Physical punishment, including lack of sleep or food.

None of these factors by themselves are dispositive, but some may be more important than others.

Although the Constitution does not require proof of knowledge of a right to refuse as the *sine qua non* of an effective consent to a search, such knowledge was highly relevant to the determination that there had been consent.

United States v. Mendenhall, 446 U.S. 544, 558-59 (1980).

Sometimes an officer will tell a defendant that she/he will ask for a search warrant if consent is not given. Most courts consider that such a scenario does not constitute coercion and consent in such circumstances is valid. *State v. Smith*, 115 Wn.2d 775, 801 P.2d 975 (1990).

In Washington there are additional limitations on consent searches when the police have gone to the to-be-searched house on a “knock and talk” mission.²⁸ In Washington, to be effective, consent obtained as a result of a knock and talk must be preceded by advising the consensor that they need not consent to the entry/search. *State v. Ferrier*, 136 Wn.2d 103, 960 P.2d 927 (1998).

iii) Consent may be limited and may be withdrawn

Consent may be limited and may be withdrawn. A search of areas for which consent was not given or after consent is withdrawn is invalid. See Model Code of Pre-Arrest Procedure § SS 240.3 (1975) - a consent search “shall not exceed, in duration or physical scope, the limits of the consent given”, *Mason v. Pulliam*, 557 F.2d 426 (5th Cir. 1977).

²⁸“Knock and talk” is generally understood to mean situations where the police go to a suspect’s house without a warrant in an attempt to get the suspect to voluntarily talk to the police and/or consent to a search.

Some courts have held searches invalid when the officers, acting under general consent, broke open locked or sealed containers. *State v. Wells*, 539 So.2d 464 (Fla. 1989). Is a search of hidden, password protected or encrypted files under a general consent to search a computer open to the same attack? To the extent that analogy to the physical world is applicable, the answer is probably no. The line of cases represented by *Wells*, focus on the destruction of property necessary to open the sealed or locked container. Since there is no comparable destruction in accessing hidden, password protected or encrypted files, the search may be upheld.

One factor that influences courts in determining the scope of a consent is whether the person who gave the consent was told of the focus of the search. If so, then containers that could contain the focus can be searched under consent. *Florida v. Jimeno*, 500 U.S. 248, 111 S.Ct. 1801, 114 L.Ed.2d 297 (1991). If consent is given when no focus is delineated, courts interpret the consent more generally. *United States v. Snow*, 44 F.3d 133 (2d Cir. 1995).

b) Search incident to arrest

When an arrest is made, officers are allowed to search the person arrested for any weapons or for any evidence on the arrestee's person in order to prevent its concealment or destruction. The officer may also search areas within the arrestee's immediate control for the same purpose.

Chimel v. California, 395 U.S. 752, 762-63 (1969).

This doctrine would allow an officer to look for computers or computer evidence within the arrestee's immediate control but would not allow them to further search the computers without other legal justification.²⁹

c) Exigent circumstances

The exigent circumstances exception allows a warrantless search of premises where the officer has probable cause and exigent circumstances exist. The determination of whether exigent circumstances exist includes an examination of many factors. (See *United States v. Brown*, 52 F.2d 415, 421 (2d Cir. 1995) and *United States v. Rico*, 51 F.3d 495, 501 (5th Cir. 1995) for examples of some of these.)

Analysis of exigent circumstances in computer based evidence cases will likely follow how such circumstances are analyzed in narcotics cases where there is a belief that contraband is being destroyed.

We have long recognized that the imminent destruction of evidence may constitute an exigency excusing the failure to procure a warrant. This risk

²⁹In an unreported decision (*State v. Felton*, No. 47773-1-I, 1/28/2002) Division I addressed a case where the suspect was arrested on suspicion of auto theft. In a search incident to arrest the police discovered a computer in the suspect's bag. The police suspected the computer was stolen and accessed the files to locate information about the owner. They contacted the person whose name was found in the computer who told them the computer was indeed stolen. Division I suppressed the evidence obtained from the computer including the information from the owner and reversed the conviction, holding that although the computer was discovered pursuant to a proper search incident to a lawful arrest, and although the police had probable cause to believe the computer was stolen, they were not authorized to conduct a search of the computer's files without a warrant. The court reaffirmed Washington's automatic standing rule.

is particularly weighty where narcotics are involved, for it is commonly known that narcotics can be easily and quickly destroyed. . . We held that a police officer can show an objectively reasonable belief that contraband is being, or will be, destroyed within a home if he can show 1) a reasonable belief that third persons are inside a private dwelling, and 2) a reasonable belief that these third persons are aware of an investigatory stop or arrest of a confederate outside the premises, so that they might see a need to destroy evidence.

United States v. Dawkins, 17 F.2d 399, 405 (D.C. Cir. 1994).

In the context of computer searches, the important thing to understand about this exception is, like a search incident to arrest, it does not justify a search of computer. It might allow for a search for and seizure of a computer to preserve evidence under the right facts, but a separate legal justification to search the computer must be provided.

d) Inventory

While inventory searches are generally thought of as searches of vehicles, one type of an inventory search that might apply to computers is a booking search. See generally *Illinois v. Lafayette*, 462 U.S. 640, 648 (1983) and *United States v. Thomas*, 11 F.3d 620, 628 (6th Cir. 1983).

While such a search, like a search incident to arrest or an exigent circumstances search, is probably limited to removing a computer (such as a laptop or other portable computing device) from a suspect's person as opposed to searching the computer itself, in some circumstances a search of the computer to ascertain or verify a defendant's identity may be appropriate. See *Illinois v. Lafayette, supra*, at p. 646 for a discussion of this issue in general.

e) Stop and frisk

[A] law enforcement officer, for his own protection and safety, may conduct a patdown to find weapons that he reasonably believes or suspects are then in the possession of the person he has accosted.

Ybarra v. Illinois, 444 U.S. 85, 93 (1979)

While any computing device found during such a frisk which is contraband may be seized, this exception to the warrant requirement is generally not applicable to computer based evidence search cases.

f) Mobility

An officer with probable cause to search a vehicle he/she has lawfully stopped may conduct such a search without a warrant because the automobile's ready mobility constitutes an exigency sufficient to excuse failure to obtain a search warrant. *LaBron v. Kilgore*, 518 U.S. 938, 940 (1996). There may be a case where probable cause exists to search a vehicle for an item which could be found in a computer, thus allowing a warrantless search of a computer under this doctrine but there are no reported decisions involving such a case. As stated earlier, there is little likelihood that a laptop computer could be searched without a warrant under this justification.

g) Plain view

The rationale of the plain view doctrine is that if contraband is left in open view and is observed by a police officer from a lawful vantage point, there has been no invasion of a legitimate expectation of privacy and thus no search within the meaning of the Fourth Amendment - or at least no search independent of the initial intrusion that gave the officers their vantage point.

Minnesota v. Dickerson, 508 U.S. 355, 375 (1993).

This exception has three rules when applied in the computer evidence search context.

- [The incriminating nature of the item in plain view must be immediately apparent;
- [The officer must be lawfully located in a position from which he or she can plainly see the item;
- [The officer must not change the focus of her/his search as a result of discovering the plain view item.

The first two of these rules come from the development of the plain view doctrine in the non-computer context. *United States v. Bradshaw*, 102 F.3d 204, 211 (6th Cir. 1996). The third comes from the unique aspects of a search for computerized evidence.

For example, if a search protocol calls for examining all files sequentially on a computer, then every file may be examined.³⁰ If incriminating evidence outside the scope of the warrant is observed in one file (such as child pornography) and then, still following the search protocol, other files are examined which also contain incriminating evidence outside the scope of the warrant, all qualify as plain view evidence.

On the other hand, however, if the search is for records of business transactions and the search protocol calls for a key word search and an examination of the directory and file structure, but not the examination of each individual file, then the discovery of a possibly incriminating file name (“babyfuk.jpg”, for example) does not justify the opening of that file. Furthermore the discovery of child pornography, in plain view and outside the scope of the warrant, does not justify a change in the search protocol or search focus. *United States v. Carey*, 172 F.3d 1268 (10th Cir. 1999).³¹

With a solid search protocol, the police can follow that protocol, even if they discover

³⁰This and subsequent examples assume the officer has a valid warrant allowing her/him to be at the location, with the object they are authorized to seize being such that it could be found in a computer.

³¹The defendant, who was under investigation for narcotics violations, consented to a search of his apartment. Two computers were seized and a warrant was issued allowing the computers to be searched for information relating to the sale and distribution of controlled substances. In conducting this search of the computer the police discovered child pornography. The officer testified that while he did not originally believe the jpg files he examined contained child pornography, after he saw the first one he had probable cause to believe the others contained child pornography. He continued to examine the jpg files, finding more child pornography. He did this examination on another computer because the files could not be viewed on the suspect’s computer.

The court interpreted this continued examination of the jpg files as an expansion of the scope of the search to include items outside the warrant’s parameters.

incriminating evidence outside the scope of the warrant, and be confident that the evidence will be admissible. Without such a protocol, or if the new evidence creates the need to go outside the protocol, the best practice when outside-the-scope-of-the-warrant evidence is discovered is to stop the search and get a new warrant based on the plain view evidence discovered.

B. Statutes

1) Privacy Protection Act

The Privacy Protection Act, 42 U.S.C. § 2000aa (PPA), was passed in response to *Zurcher v. Stanford Daily*, 436 U.S. 547 (1978). In that case officers executed a search warrant at the offices of the Stanford Daily, a student newspaper, to search for pictures of a demonstration. The newspaper and some of its staff brought a 42 U.S.C. § 1983 action against the officer who conducted the search and others. The Supreme Court, in *Zurcher*, held that other than applying warrant requirements with particular exactitude, there were no limitations on warranted searches where First Amendment interests are involved.

The PPA, passed in response to *Zurcher*, imposed limitations. It prohibits searches and seizures of material that an individual intends to publish or broadcast and “documentary material” possessed by a person in connection with a purpose to publish or broadcast the material. Such material must be obtained by subpoena.

There are exceptions.

- [The law does not apply to material that constitutes contraband, fruits of a crime, or property designed or used to commit a crime.
- [It doesn’t apply to searches and seizures needed to prevent imminent death or injury;
- [It doesn’t apply when the material is possessed by the target of the investigation;³²
- [It doesn’t apply to child pornography.

In addition, for “documentary material” the law does not apply if there is a reason to believe that the possessor of the material would destroy, alter or conceal the evidence if a subpoena was issued, or if there is a finding that the materials have not been produced as directed under a subpoena and further delays would threaten the interests of justice.

The act only applies when there is a reasonable belief of the subject’s intent to publish the material.

Most law enforcement professionals who did computer searches were unaware of this act until the Steve Jackson Games case. In March, 1990, Secret Service agents executed a search warrant at the offices of Steve Jackson Games, Inc. (SJG), searching for a computerized text file containing information about Bell South’s emergency call system. SJG published books, magazines, role-playing games and other related products. SJG also hosted an electronic bulletin board system which, among other things, allowed customers to send and receive private email. Among the items seized in the search was the computer that operated the Bulletin Board System. There were 162 items of unread, private email on this system.

³²In an exception to the exception, this exception does not apply when the alleged crime is the receipt, communication, or withholding of such evidence unless the material relates to national defense or is classified.

In 1991 SJG filed suit against the Secret Service claiming violations of the Privacy Protection Act and the Electronic Communications Privacy Act (discussed in the following section.) *Steve Jackson Games, Inc. v. United States Secret Service*, 816 F.Supp. 432 (W.D. Tex. 1993). The trial court held that the Secret Service violated the PPA and awarded damages of \$51,040. The court also found a violation of the ECPA and awarded statutory damages of \$1,000 to each individual plaintiff. It also awarded SJG over \$250,000 in attorney fees and costs.

The court found that although the Secret Service may not have known that SJG was a publisher at the time of execution of the warrant, they were aware of this by the day after the execution of the warrant. Although SJG asked for the prompt return of the material seized, and, the court found, the material could have been copied and returned “within a period of hours and not more than eight days”, the material was not returned for several months. It was the failure to return the seized property that exposed the Secret Service to liability.

The decision makes several things clear. First, a computer “publisher” is protected under the PPA. Second, even if there is no knowledge of publisher status at the time a search is executed, if the seized material shows publisher status, the seizing agents may be liable for their actions following their presumed acquisition of knowledge of publisher status. Finally, SJG was not itself a target in the investigation. Had the company been the target, the results might have been different.

The decision also makes it clear that a lack of familiarity with the Privacy Protection Act can be costly.

The most important point about the PPA that can be made in a summary discussion such as this one is that the Act applies almost exclusively to third party search situations. When you seek to search your target’s computer you are usually (but not always) outside the parameters of the PPA. Subsequent decisions to *Steve Jackson Games* suggest that the PPA will not be applied broadly where the suspect’s computer is involved. In *Guest, et al v. Lies, et al*, 2001 WL 733419 (6th Cir. 7/2/2001) the Sixth Circuit decided a civil case brought, *inter alia*, under the PPA, involving the search and seizure of computer bulletin board systems containing obscene material. The court held that

[W]hen protected materials are commingled on a criminal suspect’s computer with criminal evidence that is unprotected by the act, we will not find liability under the PPA for seizure of the PPA-protected materials. We emphasize, though, that police may not then search the PPA-protected materials that were seized incidentally to the criminal evidence. [citation omitted]

a) Suppression

A violation of the Privacy Protection Act is not a basis for suppressing the evidence obtained. “[E]vidence otherwise admissible in a proceeding shall not be excluded on the basis of a violation of this chapter.” 42 U.S.C. § 2000aa-6.³³

³³In *Oklahoma v. One Pioneer CD-Rom Changer*, 891 P.2d 600 (Okla. App. 1994), the state sought to forfeit computer equipment used in the distribution of pornography. The system owner claimed a violation of the PPA because the material seized contained non-offending material that was intended for publication on CDs. The court rejected this defense, noting that even if the PPA had been violated, this did not preclude a forfeiture so long as the equipment was seized pursuant to an otherwise lawful search and seizure.

b) Civil liability

Civil liability can result from a violation of the Privacy Protection Act. The PPA states that an aggrieved person has a cause of action

(1) against the United States, against a State which has waived its sovereign immunity under the Constitution to a claim for damages resulting from a violation of this chapter, or against any other governmental unit, all of which shall be liable for violations of this chapter by their officers or employees while acting within the scope or under color of their office or employment; and

(2) against an officer or employee of a State who has violated this chapter while acting within the scope of or under color of his office or employment, if such State has not waived its sovereign immunity as provided in paragraph (1).

42 U.S.C. § 2000aa-6(a)

It was this provision that provided the legal basis for the liability found in the Steve Jackson Games case discussed above.³⁴

c) Good faith

Good faith reliance on an otherwise valid search warrant is a defense against civil liability under the PPA. In *Davis v. Gracey*, 111 F.3d 1472 (10th Cir. 1997), the plaintiff sued law enforcement officers and entities involved in a search and seizure of a computer system involved in the distribution of pornography. The plaintiff contended the police seized materials that the defendant intended to publish on a CD, thus violating the PPA. The district court granted summary judgment to the defendants, holding that the defendants were entitled to the good faith defense of 42 U.S.C. § 2000aa-b(b) because the police had acted in reliance on a valid warrant.³⁵

³⁴Liability was also found under the ECPA, discussed below.

³⁵On appeal the Tenth Circuit dismissed the suit because of an absence of subject matter jurisdiction. All of the entities had been previously dismissed leaving only the individual officers. The Court held that the PPA only allowed actions against governmental entities unless the state had not waived sovereign immunity. Since Oklahoma had waived immunity, only the government, not the individual officers could be sued.

2) Electronic Communications Privacy Act

The Electronic Communications Privacy Act (ECPA) updated the wiretap statute (18 U.S.C. §2510, et seq.) and added the Stored Electronic Communications Act (18 U.S.C. §2701 et seq. Although it is common to reference ECPA in discussing limitations on computer searches, most often it is the specific provisions of the Stored Electronic Communications Act (SECA) that are in play.

a) What does the ECPA cover?

The act covers the contents of electronic communications while in electronic storage by an “electronic communications service.”

- [An electronic communications service is “any service which provides to the users thereof the ability to send or receive wire or electronic communications.” 18 U.S.C. § 2510(15).

The act also covers the contents of electronic communications in a “remote computing service.”

- [A remote computing service is one that provides “to the public . . . computer storage or processing service by means of an electronic communications system.” 18 U.S.C. § 2711(2)

b) What does the ECPA prohibit?

The act prohibits providers of electronic communication services or remote computing services from disclosing contents of electronic communications or customer records and transactional data to the government, except as provided.

c) When is a disclosure to the government allowed?

i) Subscriber or customer records

Subscriber or customer records (name, address, telephone toll billing records, telephone number or other subscriber number or identity, and length and type of service for a subscriber or customer) may be obtained by subpoena, warrant or court order.

The USA Patriot Act made other subscriber/customer records, such as transaction history, IP logs, etc. available by subpoena. Previously they had been available only by warrant or court order.³⁶

Pursuant to the ECPA, a subpoena can be used to obtain basic subscriber information. Basic subscriber information includes customer’s name; address; length of service; means and source of payment (including any credit card or bank account number); local and long-distance telephone toll billing records; and records of session times and durations, as well as any

³⁶ One of the problems that arises in obtaining third party records arises when the third party is out of state. A Washington judge may not have authority to issue a warrant to search a location outside of Washington. Out of state recipients of subpoenas often refuse to comply with the subpoena. For warrants the best approach may be to have Washington law enforcement contact law enforcement in the area where the third party is located and obtain a search warrant under that jurisdiction’s laws. Loudon County Virginia, where AOL is located, has a full time deputy who does little but assist out-of-state law enforcement in obtaining records from AOL. For subpoena compliance the only compulsory mechanism is the Uniform Act for Securing Attendance of Out of State Witnesses.

temporarily assigned network address.

Pursuant to the ECPA, a subpoena can be used to obtain opened email from a provider if the “customer” or “subscriber” is given prior notice of the disclosure by the government. This disclosure may be delayed for up to 90 days when notice would jeopardize a pending investigation or endanger the physical safety of a person. An extension of an additional 90 days may be considered by the issuing court. Following the delay notification period, the government must give notice.

Pursuant to the ECPA, a court order, sometimes referred as to an “articulable facts order” or “§ 2703(d) court order” may be sought to obtain all other subscriber information except the content of an unopened email that has been stored for 180 days or less. To obtain a “§ 2703(d) court order,” there must be specific and articulable facts showing that there are reasonable grounds to believe that the specified records are relevant and material to an ongoing criminal investigation. These records would include complete audit trails/logs, Web sites visited, identities of email correspondents, cell site data from cellular/PCS carriers, and opened email. As a practical matter, a “§ 2703 (d) court order” can also be used to obtain basic subscriber information. Notice to the subscriber is only required when opened email is requested from the provider.

ii) Contents of electronic communications

In general, contents may be disclosed with the consent of the parties to the communication (only one party need consent).

A government entity can compel the production of the contents of stored electronic communications by warrant. For communications in storage for more than 180 days these can be obtained by subpoena.³⁷

d) What is the remedy for violation of this act?

i) Criminal liability

The ECPA provides for criminal liability for one who:

(1) intentionally accesses without authorization a facility through which an electronic communication service is provided; or

(2) intentionally exceeds an authorization to access that facility; and thereby obtains, alters, or prevents authorized access to a wire or electronic communication while it is in electronic storage in such system.
18 U.S.C. § 2701(a)(1) and (2).

Punishment can be up to two years imprisonment.

ii) Civil liability

The act creates a civil cause of action for violations, including damages and attorney’s fees (punitive damages if the violation was willful.) Although one decision held that only the service provider, not the government, was civilly liable for violations (*Tucker v. Waddell*, 83 F.3d

³⁷There are some differences between how the law treats a provider of electronic communications services and a provider of remote computing service.

688 (4th Cir. 1996), a later decision found liability in the government official who obtained the information. *McVeigh v. Cohen*, 983 F.Supp. 215 (D.D.C. 1998).³⁸

iii) Suppression

Suppression is not a remedy for non-constitutional violations of ECPA. 18 U.S.C. § 2708.

iv) Good faith

§ 2707(e) of the ECPA states that “reliance on . . . a court warrant or order, a grand jury subpoena, a legislative authorization, or a statutory authorization . . . is a complete defense to any civil or criminal action brought under this chapter.”

This provision was used by the Tenth Circuit in *Davis v. Gracey*, 111 F.3d 1472 (10th Cir. 1997) to reject the defendants’ civil claim under ECPA. The court noted that because the officers relied on a warrant supported by probable cause they were entitled to a “good faith” defense. *Davis*, 111 F.3d at 1484.

The Ninth Circuit has made it clear, however, that legal process which “transparently and egregiously” violates the relevant rules can expose the party issuing the process to liability under the ECPA. *Theofel v. Farey-Jones*, 341 F.3d 978 (9th Cir. 2003).

3) Washington Privacy Act (RCW 9.73)

Washington’s Privacy Act has been called one of the most restrictive in the nation.³⁹ While the Act has historically been applied in the context of aural communications, a recent case states clearly that electronic communications are protected by the act as well.⁴⁰

The devices used to capture parts of these communications – pen registers and trap and trace devices – were, until 1998, treated differently.

In *State v. Gunwall*,⁴¹ the Supreme Court held that the information obtained by a pen register – the numbers called from a telephone including the date and time, but not the contents of the call – was a private communication, entitled to protection under the Privacy Act. In *State v. Riley*⁴² the Court held that the information gathered by a trap and trace device – the origination of calls to a particular, monitored location – were not private communications and not covered by the Privacy Act.

In 1998, the Privacy Act was amended to add section 9.73.260 dealing specifically with pen registers and trap and trace devices. Section 260 eliminates the case law distinction between the two devices – both are prohibited by the Privacy Act unless specific procedures are followed.

It is clear under this section that if the procedures are followed, a pen register and/or a

³⁸The different outcomes in these two cases may be explained by the different sections of the ECPA under which the plaintiffs sued. In *Tucker*, where the plaintiff sued under 18 U.S.C. § 2703(c) (which imposes limits on service providers) the court found no liability on the behalf of the government. In *McVeigh*, where the plaintiff also sued under 18 U.S.C. § 2703(a) and (b) (which impose limits on the government) the court found government liability.

³⁹*State v. O’Neill*, 103 Wn.2d 853 (1985).

⁴⁰*State v. Townsend*, at n. 1 above.

⁴¹106 Wn.2d 54, 720 P.2d 808 (1986).

⁴²121 Wn.2d 22, 846 P.2d 1365 (1993).

trap and trace device may be installed on a traditional telephone line. But may such devices be used to monitor the source and destination of electronic communications like e-mail? A surface glance at the statute reveals some apparent inconsistencies on this point. A pen register is defined as “a device that records or decodes electronic or other impulses that identify the numbers dialed or otherwise transmitted on the telephone line to which such device is attached”⁴³ Is e-mail header information a “number dialed or otherwise transmitted?” Is this information transmitted on a “telephone line”?

A trap and trace device is defined as “a device that captures the incoming electronic or other impulses that identify the originating number of an instrument or device from which a wire or electronic communication is transmitted.”⁴⁴ By specifically referencing an electronic communication, this definition appears to contemplate using such a device to harvest originating e-mail header information.

While portions of the statute dealing with the contents of the order appear to be focused on traditional telephone lines,⁴⁵ other portions of the statute contemplate the service of such an order on, among others, providers of electronic communications service.⁴⁶ In the past federal law enforcement officers and prosecutors used the identical provisions of Title 18, United States Code, §§ 3123, 3127, to obtain pen register and trap and trace orders for electronic communications. There are no recorded opinions where such orders were challenged.⁴⁷

The Patriot Act changed federal law to make it clear that pen/trap devices can be used for electronic communications. Although similar changes were proposed for the Washington Privacy Act they were not passed.

All of this suggests that §260 of the Privacy Act authorizes the use of pen registers and trap and trace devices for electronic communications provided the provisions of the statute are followed.

C. Court Rules and Magistrate-imposed Restrictions

1) Court rules: Time within which search must be completed

Some jurisdictions have statutes or court rules that require a search warrant to be executed within a specified period of time after it was issued. Some courts hold a violation of such statutes or court rules to require suppression. *Sgro v. United States*, 287 U.S. 206, 53 S.Ct. 138, 77 L.Ed. 260 (1932). Other courts have held that absent a Fourth Amendment violation (i.e., probable cause still existed when the warrant was executed) suppression is not appropriate even when the statute or court rule time limit was exceeded. *United States v. Gerber*, 994 F.2d 1556 (11th Cir. 1993).

⁴³RCW 9.73.260(1)(d)

⁴⁴RCW 9.73.260(1)(e)

⁴⁵The order is to identify the person to whom the telephone line is leased or in whose name it is listed (RCW 9.73.260(4)(a)) and the number and, if known, the physical location of the telephone line to which the device is to be attached, and in the case of a trap and trace device, the geographic limits of the order (RCW 9.73.260(4)(c)).

⁴⁶RCW 9.73.260(5).

⁴⁷The federal statute was amended by the USA Patriot Act to make it explicit that pen registers and trap and trace devices can be used for electronic communications. An attempt in 2002 to modify the Washington statute to comply with the federal statute was unsuccessful.

In Washington the Criminal Rules require the search to be completed within 10 days of the issuance of the warrant.⁴⁸

In the context of computer searches the more problematic issue relates to the off-site search of a computer following the execution of the warrant. It is important to understand that even if your application justifies and the warrant approves an off-site search of a computer, the off-site search of the computer must be done within the same time limits set for the search of the premises where the computer was located. Here's why.

If a computer is contraband or an instrumentality, it may be seized. If it is a container of evidence it may be "seized" solely for the purpose of searching it off-site. There is no other legal basis for retaining the computer. Thus, if the warrant sets a three day deadline on the execution of the warrant, and the computer is not searched within that three day period, arguably there is no longer any justification for police to hold the computer and there is no authority for a search outside that three day period.

Even if the issuing judge is willing to allow additional time to search a computer off-site, the court rules discussed earlier may be an absolute limit on the judge's authority to extend that time.

Some argue that the justification for magistrate imposed and statute/rule imposed deadlines is to insure that a warrant will be executed before the information on which it is based becomes stale. Since the computer was seized during the required time and since the information in that computer will not change between that seizure and the ultimate search of the computer, there is no staleness issue. This is a fine argument in principle and may be successful in many cases. If, however, you have a court that considers the magistrate/statute/rule deadline to be absolute, then this argument may not be successful.

In *United States v. Syphers*, ___ F.Supp. 2d ___ 2003 WL 22966910 (D.N.H. December 18, 2003) the trial judge rejected a suppression motion based on a seven month's delay in analyzing computer evidence seized under a warrant. Analogizing to the seizure of suspected heroin (which is not lab analyzed for several months) the trial judge held there is no Fourth Amendment violation where probable cause has not dissipated because the item previously seized has been in control of the police since the seizure.

The thrust of this decision is that while Fed.R.Crim.P. 41(e)(2)(A) mandates the execution of federal warrants within a specified time no longer than 10 days, this rule should not be read to impose a ten-day deadline to analyze evidence already seized. If police seize evidence consistent with Rule 41 and the Fourth Amendment, any delay in the analysis of such evidence (whether it be a hard drive or a filing cabinet full of documents) is not of constitutional concern.

Nevertheless, where a computer has been seized solely as a container of evidence to be searched offsite, it begs the question to claim that the subsequent examination is an "analysis" not a search. If the only justification for seizing the computer was to search it offsite, there is a strong argument that the search has not occurred until the computer is examined.

In a Washington case involving bank records the detective obtained a search warrant for bank records and served it on the bank the same day. Consistent with CrR 2.3(c) the warrant directed that the premises be searched within ten days. The bank provided the documents

⁴⁸CrR 2.3(c).

covered by the warrant 17 days later.

The defendant objected to the admission of the bank records, arguing that they were obtained outside the ten day limit of the warrant and CrR 2.3(c).

The Court of Appeals first noted that the constitutional requirement that a warrant be served before the probable cause becomes stale was met in this case. The court held that the service of the warrant on the bank initiated the bank retrieval process and “involved the records of a disinterested business entity whose daily operations involve the creation, storage and retrieval of the records themselves. In these circumstances, service of the warrant begins the search. Therefore, the search began before the warrant expired, and so long as probable cause continues to exist through completion of the search, the search is constitutionally timely.” *State v. Kern*, 81 Wn. App. 308, 312 (1996).

The court then addressed the specific requirement of CrR 2.3(c) and held that this was concerned only with the staleness of probable cause issue, already decided. The admission of the evidence was affirmed.

What I advise the police to do is to make some search of the computer within the prescribed time sufficient to identify some contraband on the computer. Once that is done you are no longer holding the computer solely as a container which might contain evidence and which you will search in the future. You are now holding the computer as a container of actual identified evidence. Any subsequent review of the computer will be separating the wheat from the chaff - separating the evidence from non-evidence. If you do not find evidence within that time period your choices are to request an extension (which may not be effective under the court rule) or, if you continue to have probable cause, request a new warrant. At the very least you should initiate the search within the required time period.

Both of these arguments are stopgap measures until some change is made that recognizes the reality of the length of time it takes to search multi-gigabyte drives.

2) Magistrate-imposed restrictions

The magistrate issuing the warrant may impose deadlines for the search of computers seized during the execution of the warrant that are different from the date by which the warrant itself must be executed. If those deadlines are not complied, with the evidence may be suppressed. In *United States v. Brunnette*, 76 F.Supp.2d 30 (D.Me. 1999) the warrant was approved on February 4 and executed on February 9, within the time set by the warrant. The government requested 30 days to examine the computer seized in the warrant, and subsequently requested a 30-day extension. The magistrate granted both requests. Notwithstanding this deadline of April 8 to examine the computer, the search of the computer was not started until April 10. Finding no good reason for this delay, the court suppressed the evidence from the search of the computer.

Analysis of Evidence

1. At the Search Location

It is beyond the scope of this paper to discuss the details of what should be done at the search location. Other sources should be referenced for this information.⁴⁹ I will touch on a few legal issues.

A. Who Should Accompany the Officers Executing the Search?

1) Should an expert accompany the officers?

At the federal level 18 U.S.C. § 3105 precludes anyone other than the officer being present at the search except for a person “in aid of the officer.” The statute has been interpreted to allow experts, specifically provided for in the warrant, to accompany and participate in the search. *United States v. Schwimmer*, 692 F.Supp. 119 (E.D.N.Y. 1988). In *Wilson v. Layne*,⁵⁰ the Supreme Court noted that it is well established that the presence of third parties for purposes related specifically to the execution of the warrant is appropriate even when not specifically authorized by the warrant. Other courts have made it clear that such persons are there to assist the peace officer executing the search.⁵¹

A somewhat different issue may arise where computers have been taken off-site for searching and that search is conducted by an expert who is not a peace officer. Although I am unaware of any cases bearing directly on this, it is my recommendation that, in the application, you justify and request permission for this person to examine the computer under peace officer supervision (i.e., the officer will be instructing the expert in what to look for), and that permission for this activity be specifically provided for in the warrant.⁵² It is important to understand that this requesting and receiving permission process is dictated by the fact that the off-site examination is a continuation of the execution of the warrant, not merely the evaluation of already legitimately seized information where no special permission is needed.

This suggests that any time the presence of an expert will be useful (and an expert is available, of course) you should justify and request that the expert be allowed to accompany the officers and participate in the search.

2) Should a representative of the victim accompany the officers?

But what if the only expert is a representative of the victim?

Justifying the presence of an independent expert is merely a matter of explaining why an expert’s assistance is needed. Justifying the presence of a representative of the victim is a more difficult task, and, even if approved in the warrant, the presence of a victim representative always

⁴⁹*High-Technology Crime, Investigating Cases Involving Computers*, Kenneth Rosenblatt (KSK Publications 1995), *Investigating Computer Crime*, Franklin Clark & Kin Diliberto (CRC Press 1996), http://www.ustreas.gov/usss/index.htm?electronic_evidence.htm&1

⁵⁰526 U.S. 603, 119 S.Ct. ___, 143 L. Ed.2d 818, (1999).

⁵¹*Commonwealth v. Sbordone*, 424 Mass. 802, 678 N.E.2d 1184 (1997), *United States v. Clouston*, 623 F.2d 485 (6th Cir. 1980).

⁵²If you decide, after the search, to use a non-peace office expert to examine the computer, I recommend you file a supplemental affidavit with the warrant judge requesting permission for that person to search the drive.

creates a credibility issue.

The strongest argument in favor of having a victim representative present is that they can quickly identify specific items covered by the warrant and thus minimize the intrusion and inconvenience to the suspect. (Of course, the victim representative's unique familiarity and ability to recognize offending material is also the reason why their presence is particularly invasive.)

If you do request the presence of a victim representative, you must explain why only that person or other victim representative can do the identification necessary. You should stress that the person's ability to recognize offending evidence on sight will likely reduce the extent and time of the search.

The officers should be advised that notwithstanding this permission, they are the ones conducting the search, not the victim. To the extent possible the role of the victim representative should be clearly delineated in the application.

The problems associated with allowing a victim to accompany warrant executing officers is illustrated by a recent case in King County, *In re U.S. Computer Corporation*, 98-118 (Search Warrant). In that case the trial judge, William Downing, discussed some of these problems:

With the determinative issues before the court now fully resolved, any further discussion of the issues raised is certainly dicta. Fully aware of the shortcomings and limited value of dicta, this court would nonetheless feel remiss in its duties if it did not take this opportunity to give voice to its strong misgivings about the execution of the search warrant it signed on August 6, 1998.

That warrant was a court order directed to "any peace officer in the state of Washington". A "peace officer" is "a duly appointed city, county or state law enforcement officer". RCW 9A.04.110(15).

The notion of peace officers as public servants has been with us for some time and deliberately so. In Great Britain, the harsh lessons learned from a system of private law enforcement are said to have led to Sir Robert Peel's creation of the first salaried, full-time police force in 1829.

In this country as well, frontier justice, vigilantism, and lynchings live in our history as sad reminders of the need for organized society to always provide adequate, public law enforcement. Accountable to the full citizenry from which its powers derive, a police agency knows it must scrupulously follow the procedural dictates of the law. Extensive training is now given to police officers (at the expense of the public they will serve) and a good officer who does not cut corners is said to do things "by the book".

Where private law enforcement is involved, there is no "book" but only loose pages. While many of these pages may spring to mind, one apt example is this verse from the Woody Guthrie song "Buffalo Skinners" describing the approach of some earlier westerners who felt cheated in their business dealings:

*Well, those cowboys never had heard
Such a thing as a bankrupt law;
So we left that drover's bones to bleach
On the plains of the buffalo.*

It is without question that private entities should be encouraged to support public law enforcement through a variety of means. The insurance industry does so in valuable ways and various reward programs have been quite successful. Private entities' financial support for a law enforcement effort is not, by itself, tantamount to private entities seeking to perform the law enforcement function.

What is called for, it would seem, is that this public/private relationship remain somewhat at arm's length. Some distance is essential both to the police agency's ability to maintain its loyalty solely to the broad public and also to that public's confidence that such is the case. Although it may sound Pollyannaish, the theory is that the police officer works at once for the suspect as well as the victim. A private security officer serves but one master.

If, for example, company "A" were to suspect itself to be being victimized by company "B" through illegal means, it would logically be motivated by a strong desire to gather confirmation. One approach would be to file a civil lawsuit and utilize the discovery process in hopes of gaining information. A second alternative would be for A's security personnel to travel to B's headquarters and announce "We'd like to come in and poke around a little bit and also ask your employees a few questions about your illegal activities."

The first alternative is slow and laborious and subject to obstructionism. The second is laughable in its chances of success; acting on their own, the security personnel would never get across the rival's threshold.

In this case, it was entirely appropriate for Hewlett-Packard security personnel to accompany law enforcement officials as they (the law enforcement officials) served this court's warrant to search USCC. A role for the "victim" in helping to identify stolen property for which there is probable cause to search, was contemplated by this court and is fully consistent with established law. See, e.g., *Wilson v. Layne*, 526 U.S. 603 (1999).

The role played by such a citizen participant in the service of a search warrant must be narrowly circumscribed. To the extent that his role is expressly or impliedly authorized in the warrant, he is acting under color of state law. Any actions beyond that, however, constitute "a frolic of his own" and cannot be countenanced.

In the service of this search warrant, the role for citizen participants that was impliedly authorized was to aid in identifying stolen memory modules

bearing fictitious or no serial numbers. Actions beyond that were not authorized.

When private citizens necessarily accompany law enforcement personnel on the execution of a search warrant, it must be expected that the police officers – familiar with the requirements of the law – will ensure that the civilians do not engage in any unlawful conduct. It may very well be that the close relationship (financial and otherwise) between the entities involved in this search provides some explanation for the blurring of what should have been more clearly defined roles.

B. What Can be Seized

This discussion is limited to situations where you are seizing computers as containers of evidence. If the warrant allows seizing a computer as an instrumentality or contraband, this discussion does not apply. So long as you have a valid basis for an off-site search, storage devices that could contain the evidence you are authorized to seize may, themselves, be seized for searching off-site.

But what about all the other computer “stuff” at the search location. Can you seize the CPU box (as opposed to removing the hard drives from the box)? The monitor? Printers? Cables and cords? Manuals?

The general rule is that you can request and seize those items necessary to achieve the core purpose of your search. If you seize floppies for searching off site, then you may seize floppy drives. The same rationale applies to other storage media that require particular devices to be useable - tape drives for tapes, DCD or CD players for DVD or CD media, etc.

A hard drive contains both the media and the device to read the information. May you nevertheless seize the CPU box and its contents - cards, memory, cables, etc? I am not aware of any decisions answering this question, but my answer is yes.

On one level you can seize the box and its contents to show that the defendant had an operating computer, a computer capable of connecting to the Internet, a computer capable of displaying a certain quality of video image, etc.

On a more fundamental level, you do not search a hard drive by examining it with a magnifying glass. You search it by having it connected to an operating computer system, including a monitor. While you can search the drive by removing it from the suspect’s system and inserting it into another system you are not required to do this. So long as you can justify why an object is necessary to achieve the core purpose of your search you can seize that object. Manuals and other instructional material, and cables and cords connected to the computer would also be covered by this rationale.

Printers, scanners, digital cameras, other peripheral devices? The answer is less clear. If you have an incriminating printout that you want to tie to a particular printer, then you may have a basis to seize the printer. If you want to show the suspect had the capability to input images into a computer, you may have a basis for seizing a scanner or a digital camera. If those items might contain images covered by the warrant they can be seized for off-site search.

C. When to Get an Additional Warrant

The simple answer is you get an additional warrant when you have probable cause to believe that evidence, not covered by your current warrant, may be found in a computer.

In fact, as discussed in the plain view section earlier, even if you develop probable cause to believe that evidence outside the scope of your warrant will be found in the computer you are searching, you need not get an additional warrant to take note of such additional evidence, so long as you follow your search protocol.

The safest course, however, is to request an additional warrant for the new information. You can do that by stopping your valid search to get the new warrant, or by completing the search authorized by the warrant, and then applying for a new warrant to complete the search with the new category of evidence as your focus.

2. Offsite/Detailed Analysis

Even more important than having an expert at the search scene is having an expert available to examine a seized computer. If there is a capable, trained and available police computer forensics expert, that is your first choice. Often there is no such person or that person is busy on other cases. The next best choice is an independent investigator, paid for by the police. Much further down the choice list is a representative of the victim or an expert provided or paid for by the victim. The last option is to examine the data without an expert. To do so, however, runs the risk of damaging or altering data, making evidence inadmissible and overlooking key evidence.

A. Accurate Pristine Mirror

In olden days, when I first started prosecuting cases involving computer based evidence, we actually worked with the original data. This was justifiable at first. The data was on floppy drives; the write protect tab could be set on the floppy so no alterations would be made. But even when the first hard drives started appearing in our cases, we worked with the original drives.

Then the conventional wisdom became to make a backup before working with the data. That way you could always compare the original, even if modified, to the backup. We slowly moved from just making a backup to actually installing the backup on a new hard drive and doing our analysis on that drive.

Why didn't we do that from the first? The primary reasons were time, the cost of a replacement drive, and the lack of an expert who could make the backup and testify as to its accuracy. Often I was my only expert.

Making a backup, however, had a major weakness. Backups, as we think of them, back up files and directories, i.e., they recognize what the operating system recognizes as data. But there is much information on a drive that is not recognized by the file or operating systems. Deleted files are a common example. The file system does not recognize deleted data, but if the area on the drive where the file existed has not been overwritten, the data still exists.

Even if the file has been overwritten, unless the new file takes up all of the cluster allocated to the old file, remnants of the old file will be found in the part of the cluster following the new file.

Often the areas of a drive not recognized as data by the operating system - slack space, ambient space - contain the most incriminating information.

So always make a mirror image - a copy of all magnetic information on the drive.

There are several programs that have established their validity in court for making accurate backups. Among these are SafeBack by Sydex, (www.sydex.com) and Encase by Guidance Software, Inc. (www.guidancesoftware.com).

Useful tools to use in conjunction with these programs are available from New Technologies out of Portland, Oregon (www.secure-data.com).

In the ideal case I would have one mirror image made, and that image restored to four hard drives where economically feasible: One for courtroom use, one for analysis, one for the defense and one for the prosecutor.⁵³ In many cases only the courtroom and analysis copies are used. The prosecutor and defense attorney often don't want to see "how sausage is made."

B. Analysis Protocol

As seen earlier, it is important to have a protocol for analyzing the drive. A protocol is important to insure that you have exhaustively examined the drive. A protocol is important to establish the validity of your examination and the expert's report. A protocol is important to demonstrate that your examination of the computer was as minimally invasive of the owner's privacy as possible.

While the general structure of the protocol will be the same in most cases, it will almost always vary in detail from case to case. The protocol will often be a dynamic process, changing as you examine the computer and gather information.

Think of a protocol as crime scene investigation techniques. Modern law enforcement officers know not to disturb the crime scene. Modern detectives and medical examiners follow certain sets of procedures in examining a dead body. You should have similar protocols for investigating computer scenes and computers.

Examination of the directory and file structure and initial examination of only those directories and files that appear relevant on their face is an appropriate first step. Subsequent action may be based on what your initial examination reveals. For example, examination of other directories and files may be justified if the names of the directories and files you do examine do not match their contents.

Most investigators I work with use key word searches of all of the drive's binary data (obtained from the mirror image) as an initial first step. This is a useful way to determine if evidence exists (which allows you to retain the computer - recall the discussion on the time limits on the execution of warrants.) This also helps you focus in on suspect directories and files. But framing the search terms requires the assistance of someone who is familiar with the information for which you are searching - often a representative of the victim.

Having a search plan can also help insulate law enforcement against a claim of undue invasion of privacy. Although most courts view computer searches like any other container search (where you can search the entire container for what you are authorized to seize), the *Gall* case (discussed at n. 21 above) suggests that some court may be willing to afford more privacy protection to a computer than ordinary containers.

⁵³We have not yet had to test the economics of my policy. All defendants have pled guilty since I implemented this policy and we have not had to actually make multiple copies on hard drives.

In practice, however, most forensic examinations of computers cannot follow a pre-written script. What the examiner should do is keep detailed notes of what examination was undertaken and the reasons for each part of the examination. This practice was commented on favorably in the trial court's ruling on the motion to suppress in *United States v. Triumph Capital Group, Inc. et al.*, Criminal No. 3-00CR217 (EBB), <http://www.ctd.uscourts.gov/Opinions/110402.ahn.triumph.pdf> (District of Connecticut, 2002).

C. What Must be Returned and When

Occasionally this is dictated by the warrant itself. The magistrate may instruct you to complete the examination of the computer and return items not covered by the warrant within a certain amount of time. Other times the defendant or other owner of the computer may request (or demand) its return. In other situations you may want to return items not covered by the warrant on your own, particularly if this is a third party's equipment or data and you need their continued cooperation.

So, what do you do when the computer owner wants the original items returned?

There may be times when the prudent course is to return the original computer, or, more particularly, the original hard drive (or a copy), to the owner. Courts may be more inclined to view your search more generously, and to give you more leeway as a defendant in a civil case, when you have made every effort to accommodate the reasonable requests of the original computer owner. If the computer is the center of an ongoing legitimate business, even the defendant's own legitimate business, you should make efforts to return enough of the data you seized to allow the business to continue. One could argue that the Secret Service was treated more harshly in the Steve Jackson Games case because they held on to the defendant's email (crucial to the operation of his business) for several months, despite the owner's repeated requests for return of the email, and despite the email's irrelevance to the case the Secret Service was investigating.

1) Stipulation

If you are returning original evidence, whether it is just the hard drive, or the complete computer system, you should be careful to get a stipulation from the defendant, eliminating any objections as to authenticity or foundation, before returning the evidence. If the defendant refuses to so stipulate, you have a valid reason for hanging on to the original evidence. Some suggested language would be:

[suspect], in the interest of expediting the searching and seizing of records and other evidence as authorized by Search Warrant #_____, signed by _____, Judge, on _____, 2001, so as to minimize interruption of the normal computing activities of [suspect or suspect's company], stipulate to the following terms applicable to the records, equipment and evidence itemized in the attached Inventory, incorporated by reference:

[suspect] is satisfied that the backup or mirror image copies, made on _____, 2001, are complete and accurate copies of the entire contents of the systems searched as of that date. [suspect] will not contest the accuracy, reliability or source of any record copied, printed out or derived

from those backups/mirror image copies. [suspect] waives any objection as to best evidence, authenticity or foundation as to any record copied, printed out or derived from those backups/mirror image copies.

2) What if the computer owner is not the suspect?

But what if the person from whom the computer was seized, and who wants it back, is not the suspect? The suspect has no incentive to stipulate in this context and the owner of the computer can't stipulate to admissibility of evidence against the defendant. In this case try to reach a compromise with the owner. In most cases a mirror image on a different drive will be sufficient.

If the owner must have the original, and the suspect will not stipulate, so long as you have a valid mirror image and an expert who can testify as to its creation, you should be safe in returning the original even without a stipulation. The absence of a stipulation does not make your mirror image and evidence from that mirror inadmissible, it simply raises your proof hurdle.

3) Contraband on computer

Unless a computer is forfeited, there will come a time when you must return the computer to its owner. That may be early in the case when you determine there is no useful evidence on the computer or when a court orders the return of non-evidentiary items. It may occur when you have mirrored the hard drive and are returning the original under a stipulation as described earlier. It may occur when you have seized a business computer or a third party computer and wish to get it back to a non-suspect owner.

But what if there is contraband on the computer? Must you return it? Must you destroy it?

a) Obvious contraband

There are certain items that are contraband per se and should never be returned. These include, for example, stolen property (data), pirated software, and child pornography. While these should not be returned, neither should they be destroyed arbitrarily. Often the "owner" of this contraband has the right to challenge your retention of this material and you face the possibility of civil liability for destroying the items without authority. Seek permission from the judge issuing the warrant (or your own legal advisor) before deleting this material.

b) Suspected stolen property: No irreversible harm in returning

Where there is a legitimate question as to ownership of property, you may have the right to withhold the property in question until a court has determined ownership. For example, if you have what appears to be pirated software, can you withhold the software from the person you seized it from until they prove their ownership? While this may vary from case to case, I would be very careful in presuming software to be pirated and withholding it from the purported owner. In these days of buying software online, there may be no books or documentation, no specially produced CD or floppy disk, and the software is still legitimate. If you have concerns you can contact the Software Company and alert them to what you've found. They have seizure remedies available to them under the Federal Copyright Act.

c) Suspected stolen property: Irreversible harm in returning

If the property about which there is a dispute is proprietary information - information that if returned to the defendant could be copied or transmitted to others - after-the-fact remedies by the true property owner may be too late. In this instance the cautious approach is to hold on to the property until the warrant judge has directed what is to become of the property.

d) Other types of contraband: Passwords, hacking programs

Often computer criminals have cybercrime-related contraband on their computers - passwords of others, hacking programs, utilities for cracking password accounts, etc. What can/should you do with those? Passwords of others, under federal law and Washington law, are “access devices”. Access devices in the names of two or more different people besides the possessor are presumed, in Washington, to be stolen. You are entitled to treat this like any other clearly identified stolen property. You keep it and return it to the lawful owner.

Hacking programs, like burglar tools, are different. Just like many non-criminals own crowbars, many non-criminals own hacking programs. Such tools and programs serve many legitimate purposes and cannot be considered contraband per se. In addition, most hacking programs can be easily found on the Internet. Deleting such a program simply delays its subsequent acquisition by a few minutes.

Presentation of Evidence at Trial

1. Rules of Evidence and Admissibility Issues

The focus of this training CD-ROM is on obtaining and using email. For a further discussion of computer records in criminal investigations, please refer to the federal section of this CD-ROM.

There are four steps necessary for introducing email in court. First, it must be relevant. Second, it must be authenticated. Third, it must withstand a “best evidence” objection. Fourth, the contents of the email must not be inadmissible hearsay.

A. Relevance

Only relevant evidence is admissible. (ER 402.) All relevant evidence is admissible, except as limited by constitutional requirements or as otherwise provided by statute or the Evidence Rules of other rules or regulations applicable in the courts of Washington. (ER 402.) Relevant means having any tendency to make the existence of any fact that is of consequence to the determination of the action more probable or less probable than it would be without the evidence. (ER 401.) Theories of admissibility include:

Direct evidence of a crime	Credibility of witnesses
Circumstantial evidence of a crime	Impeachment
Identity of perpetrator	Negates or forecloses a defense
Intent of perpetrator	Basis of expert opinion
Motive	Lack of mistake

B. Authentication

“Authenticating” evidence means introducing sufficient evidence to sustain a finding that the writing is what you say it is. (ER 901(a).) In most cases this is accomplished by showing the writing to the witness and asking, “what is this?”

You can authenticate documents by:

- [Testimony of a witness with knowledge of the document. (ER 901(b)(1).)
- [Comparison by Court or Expert Witness. (ER 901(b)(3).)
- [Distinctive characteristics and the like. (ER 901(b)(4).)
- [Process or system. (ER 901(b)(9).)

Any other way that will sustain a finding that the writing is what you say it is since the Evidence Code does not limit the means by which a writing may be authenticated and proved. (ER 901(b).)

Common ways to authenticate email include:

- [Chain of custody following the route of the message, coupled with testimony that the alleged sender had primary access to the computer where the message originated.

- [The content of the email writing refers to matters that only the writer would have been aware.
- [Recipient used the reply function to respond to the email; the new message may include the sender's original message.
- [After receipt of the email, the sender takes action consistent with the content of the email.

In the majority of cases it is a variety of circumstantial evidence that provides the key to establishing the authorship and authenticity of a computer record. For further information, please consult the federal evidence portion of this CD-ROM.

Challenges to the authenticity of computer records often take on one of three forms. First, parties may challenge the authenticity of both computer-generated and computer-stored records by questioning whether the records were altered, manipulated, or damaged after they were created. Second, parties may question the authenticity of computer-generated records by challenging the reliability of the computer program that generated the records. Third, parties may challenge the authenticity of computer-stored records by questioning the identity of their author. For further information, please consult the federal evidence portion of this CD-ROM.

C. Best Evidence Rule

ER 1003 makes duplicates admissible to the same extent as an original unless “(1) a genuine question is raised as to the authenticity of the original or, (2) in the circumstances it would be unfair to admit the duplicate in lieu of the original.”

In a criminal action it is also necessary for the proponent of the evidence to make the original available for inspection at or before trial. For email, this is especially important, given the wealth of information contained in its electronic format, as opposed to its paper image.

Also, remember that ER 1001(c) states that the printed representation of computer data, shown to reflect the data accurately is an “original”. Thus, a printout of information will not present any “best evidence rule” issues absent a showing that the information is inaccurate or unreliable.

D. Hearsay Rule

The first question to ask is whether the information within the document is hearsay. If it is determined to be hearsay, then you need an applicable exception.

The first rule is that a printout of the results of the computer's internal operations is not hearsay evidence. These log files are computer-generated records that do not involve the same risk of observation or recall as human declarants. Thus, email header information and log files associated with an email's movement through the Internet are not hearsay. An excellent discussion on this issue can be found on the federal evidence portion of this CD-ROM.

Log files and other computer-generated records from Internet Service Providers will also easily qualify as business records. The determining issue will be whether the court finds the records reliable. *State v. Ben-Neth*, 34 Wn. App. 600 (1983). Remember, the custodian of records need not completely understand the computer or process involved in creating or maintaining the records. *Id.* Nor does the custodian need to establish the reliability of the equipment used to create or maintain the records. *State v. Kane*, 23 Wn. App. 107 (1979).

If the computer is used merely to store, analyze, or summarize material that is hearsay in nature, it will not change its hearsay nature and you will need an applicable exception for introduction. Common exceptions for the contents of email include: statement of the party, adoptive admission, statement in furtherance of a conspiracy, declaration against interest, prior inconsistent statement, past recollection recorded, business record, writing as a record of the act, or state of mind.

2. Computer Printouts and Summaries

As noted above, with proper foundation courts have routinely admitted computer printouts as evidence of computer contents. *State v. Ben-Neth*, 34 Wn.App. 600, 663 P.2d 156 (1983). Often the problem with computer printouts is not their admissibility but their volume. Evidence Rule 1006 allows summaries to be used in many circumstances. Summaries of computer records, if they meet the other requirements of the rule, as just as valid as other records.

3. Display Computer Contents on Screen/Demonstrations

Any "live" presentation of computer data is always fraught with peril. The data may not look like you want it to. The program may act differently in court than in your office. The hardware may not display the data correctly. The electrical outlets will be too far away, or not have enough plug-ins, or the outlets will not be three pronged when all your equipment is. Murphy's law applies at least as well to live presentations as it does to other aspects of our lives. Glitches show up even in Bill Gate's attempts to do live presentations or in crucial live presentations in front of Judge Jackson in the Microsoft antitrust trial.

Having said that, live demonstrations can be amazingly effective. Seeing the incriminating evidence on the defendant's own screen removes the evidence from the virtual world and makes it real and attached to the defendant. Seeing the evidence on the defendant's own screen tends to help validate your expert and your other theories of the case.

As with any live demonstration, however, it can be turned against you. If your live demonstration produces changes in the hard drive (a copy of the mirror image, of course, not the

actual hard drive) that were not found on the actual hard drive, you will have to explain why (or why not.) If the hardware you use differs at all from the defendant's, have a witness prepared to explain why the differences don't affect the evidence.

One example of the problems that come from not thinking ahead. A deputy prosecutor had the entire contents of the defendant's computer put onto a CD-ROM. (It shows how long ago this was that the CD could contain all of the hard drive.) She was preparing to demonstrate where files downloaded from news groups were stored by default on the defendant's computer. But when she started Windows and downloaded a file, it did not appear in the defendant's default location. The problem, obvious in hindsight, was that she was booting and starting Windows from her presentation laptop, and even though the CD with the defendant's copy of Windows and the settings associated with that copy of Windows was in the drive, the laptop's default settings were the ones recognized, not the defendant's. Luckily she found this out in pretrial preparation, not in the courtroom.

4. Animation

If I have learned anything as a parent, it is this: never underestimate the ability of cartoons to hold the interest of a three-year-old with an attention span of thirty seconds.

If I have learned anything as a trial lawyer it is this: never expect a jury to have an attention span longer than a three-year-old, particularly when you are presenting crucial evidence.

It takes very little insight to realize that what will hold the attention of a three-year-old will also hold the attention of a jury. Not just graphics, not just charts and pictures, but ANIMATION.

Mental images are perceived and retained far better than spoken words. With effective animation, a jury works with a visual image of the accident, the crime, and even the contract, rather than abstract concepts.

If your case turns on the jury understanding unfamiliar technology or a complex factual situation, I suggest you consider hiring an animator to present the technology or facts.

5. Experts

Guidance Software maintains a collection of cases in which the admissibility of the results of using Encase software was in issue. That collection can be found at <http://www.encase.com/corporate/legal/index.shtm>.

In a computer crime courtroom an expert serves three purposes. The first is education. The second is providing legal validity. The third is providing persuasion.

A. Expert as Educator

When the matter can be explained with just a little background information this task is easy. The problem comes when the matter to be explained is complex.

How do you explain a complex matter? For truly complex matters accept that you will not be able to explain it sufficient for all to understand. What you can do is explain it in a way that makes sense - is not counterintuitive. Where possible an explanation should have analogies to the familiar. The problem is, not all parts of the electronic or virtual world have a sufficiently close real world counterpart for analogizing to be accurate or useful. David Weinberger, a commentator on NPR, described this as “framejacking” - hi-jacking our normal frames of reference:

“We’re using age old methods of . . . reasoning trying to evaluate an unfamiliar situation by finding analogy to a familiar one But applied to the web the analogies get so complex that the familiar turns into the unfamiliar To make the situation accurate you have to make it so weird that it doesn’t help us figure things out.”

Nevertheless, analogizing from the familiar to the unfamiliar, from the simple to the complex, is a time proven method for learning new things.

B. Expert for Legal Validity

There are many requirements that must be met before evidence can be admitted. A proper foundation must be laid. Authenticity and relevance must be shown. To introduce evidence that came from a computer you searched you have to demonstrate that the evidence you seek to introduce did in fact come from the computer you searched. If you have recreated evidence from deleted or partially deleted files, you must be able to demonstrate the reliability of your recovery. Admissibility issues are typically left to the sound discretion of the trial judge. Probably the most important criteria for determining admissibility of evidence that is neither clearly admissible nor clearly inadmissible, is the believability of your expert. When the law does not provide clear guidance, judges are much more likely to allow in evidence in which they believe as opposed to evidence about which they have doubts.

C. Expert for Persuasion

Unless there is only one interpretation of the facts and no disagreement about that interpretation, after an expert educates, and after an expert qualifies evidence as admissible, the expert must persuade the fact finder that her/his version of the facts or interpretation of the facts is the correct one. Expert witnesses are a special category of witness. Unlike other witnesses, an expert can give an opinion. But that opinion must be persuasive.

An effective expert is one who is not argumentative, demonstrates no bias, and is knowledgeable but human. If the issue is truly complex then all fact finders are looking for someone they can trust to explain it to them. If your expert comes across as trustworthy then your expert will be able to persuade.

In those cases where you think the fact finder would accept your version of the case if they only had a rudimentary level of knowledge, it might be useful to have two experts - one for education and the other for legal validity and persuasion.

Unique Issues

One of the paradoxes of computer searches is that Ockham's Razor (paraphrased, "the simplest explanation is most often the correct one") is both correct and incorrect. Often, in searching computer records things are simpler than they seem. Directories and file names do describe their contents. Incriminating files are neither hidden nor encrypted. File dates and times are accurate and can be relied upon.

And often simple explanations are not correct. Date and time stamps are inaccurate because the user's clock is off, or because a difference in time zones is involved, or because the user has changed the date and time.

What I describe in this catch-all section are some small issues that may arise in computer search cases, that although small, may be enough to render the evidence suspect if the issues are not understood.

1. Dates and Times

How does the operating system date files? What does the operating system consider to be "accessing" a file? What does the operating system consider to be "creating" a file? What does the operating system consider to be "modifying" or "writing" a file? The answers to these questions are beyond the scope of this article. Information about this can be found in general books on the operating system involved. Detailed information can often be found at www.zdnet.com and other computer information sites on the Internet.

Of course, the accuracy of the date and time stamped on a file is dependent upon the system's date and time itself being accurate. The system's date and time is kept current⁵⁴ when the system is on, but when the computer is turned off the system date and time remain accurate only so long as the battery backed CMOS remains accurate. As the battery starts to wear out, the system date and time will slowly slip behind actual date and time.

If the frailties of the system itself weren't bad enough, the date and time can also be manually modified. This can be as simple as setting a different date and time for the system or the only moderately more complex act of modifying a file's own date and time stamp.

How, then, can you verify that the date and time shown as the file creation date, or last modified date, are accurate or that the system's internal date and time are accurate? Here are some suggestions:

File Accuracy:

- [Compare any dates in the content of the file with its alleged creation date
- [Compare the date and time header information in an e-mail file with the date and time stamp given that file (if it is a separate file) by the operating system

⁵⁴I learned with my own system that the date and time can be accurately stored in CMOS but get out of sync when the computer is on, if the system is overheating or otherwise causing the CPU to run slow. For several weeks earlier this year I was puzzled that my clock kept getting behind when the system was on, but immediately caught up if the system was turned off and then back on. The apparent culprit was a defective or failing fan that failed to adequately cool the CPU, causing the system itself to slow down. This is still a hypothesis, but since I replaced the fan the problem has not recurred.

- [If you have a known file date (you know when a stolen file was downloaded for example) you can compare that time to the date/time of the stolen file on the suspect's computer.

System Accuracy:

- [If the computer is attached to a network, determine if the server sets the clock of each workstation each time the user logs on;
- [Look for patterns of file creation dates and times. See if any files are dated after the date the computer was seized. If so, either the computer's internal clock is off or you're going to have a major problem explaining to the judge and jury the apparent changes to files after the seizure date;
- [Experiment. Since the date and time comes from the values stored in the computer's CMOS, not the hard drive, you need to experiment using the suspect computer BUT NOT the hard drive containing evidence. Put a new hard drive in the system, create some files and see if the file creation date matches the actual date.

You may be able to prove the actual file date/time from dates/times that are ahead or behind their actual times (because the CMOS date/time is ahead or behind.) If you can demonstrate the CMOS is two hours behind, for example, you should be able to argue that increasing the shown file date/time by two hours produces the actual time the file was created. (This is particularly true if you can independently show a particular file's date/time to be off by two hours, using one of the methods described earlier.)

In one case I did in the early 90s, *State v. Merillin Paris*, King County Superior Court No. 92-1-05560-7 SEA, we were attempting to reconstruct the sequence of events when an intruder physically went into a series of offices, turned on computers, created directories and files and copied a proprietary file. There were numerous time and date stamps on the files and directories created, there were network login times when the computers were turned on, and there were records of the security company that showed when key card access was made to the office.

My concern about whether the date/times on the files and on the log records were accurate was relieved by the system administrator's explanation and demonstration to me that on their network as configured, whenever a computer logged on to the network, that computer's clock was synchronized to the server's clock. We were able to verify that the server's clock was accurate, and thus were able to demonstrate the accuracy of the workstation clocks.

2. Authenticity

How can you prove what you present in court is what was actually on the computer you seized? On its face this seems like a chain-of-custody question. If you have witnesses who can identify the item and trace its location from when it was seized to when it was produced in court, then you can prove the item you produce is what was seized.

If all you're producing in court is the physical computer that's true. But when you're producing the contents of the computer, and claiming that those contents were on the computer when you seized it, your task is more difficult. (This is especially true if what you're introducing is evidence from the erased file or slack space of the hard drive.)

If you have a good protocol for how you handle the computer and how you recover data

from the computer, you should have no problem. But remember, any time you start up the computer from the hard drive, any time you execute files on the hard drive, any time you start up windows on the hard drive, you modify file dates. Even when there is a simple answer that does not invalidate your evidence, it's always embarrassing to have the defense point out the files with written-to dates and times after the time the computer was seized. Knowledgeable judges and juries can distinguish between file alterations done by the system that do not impact the validity of the contents or date/time stamp of other files, but don't assume you'll have either in your crucial trial.

3. How to Prove Authorship of Computer File

In some cases the existence of the incriminating file on the suspect's computer, by itself or together with other evidence you have, may be sufficient. In some cases, however, you may need to tie the evidence to the defendant beyond showing that the evidence was found on the defendant's computer.

Here are some ways you can make this connection?

- [Directory structure and location of the file within that directory structure.
- [Suspect file matches defendant's unique file-naming convention.
- [Common misspellings or grammar mistakes between suspect file and files known to have been authored by the defendant.
- [Common use of unique sentence structure, word choice, etc., between suspect file and files known to have been authored by the defendant.

4. Computer Trespass by Outsider Versus Exceeding Authority by Authorized User

Computer trespass statutes typically speak in terms of "unauthorized access" or "without authority". When the access at issue is made by someone with no authority to use the computer system there is usually little difficulty demonstrating that the access they made was unauthorized. When the access at issue was by someone who did have authority to access the system, but not all parts, or when their access is for an unauthorized purpose, proof of the "unauthorized" element of the offense can be more problematic.

Let's take three examples. All involve the computerized database of the Motor Vehicle Department of a state. In the first example, someone with no authority to access the database hacks into the system and gains access to the driving record of a candidate for political office. This is clearly unauthorized.

In the second example, an employee of the Department of Motor Vehicles whose job duties require access to the DMV computer system (for entry of payroll data, for example) but not to the driving record database, accesses the same data about the candidate. While this scenario presents some possible problems, the only real issues in this case are whether the employee knew the database was out-of-bounds, and whether the access was made intentionally. The first issue turns on what you can demonstrate the employee knew, such as the provisions in the employee manual, training materials, a screen banner before accessing the database, etc. The second issue turns on how clearly separated, the authorized and unauthorized areas are. Was the access accidental or innocent? Warning banners can be helpful here also.

The third example is the most difficult. Suppose the employee in question is one whose work directly involves the database and they are clearly authorized to access the database, but they access it for a non-business reason - to embarrass a political candidate, for example.

This example presents two distinct problems. First, if our argument is that accessing the database for an unauthorized reason is unauthorized access, then we must prove the state of mind of the suspect when they initially accessed the database. If the employee came to work, logged on her/his computer, did their work, and in the middle of the afternoon, while still logged on, obtained the candidate's records for the purpose of providing them to the candidate's opponent, is there an unauthorized access? While semantically one could argue that the unauthorized access is not the access to the database but the access to the candidate's record, most observers would consider the initial access to the database to be the "access". Thus we would have to prove, when the employee first accessed the protected database itself, it was her/his intent to get the candidate's records rather than some other legitimate task.

Second, is the real crime in this example the access to the candidate's data or the provision of that data to the candidate's opponent? Some courts take the position that the real crime in this instance is the provision of the data to the opponent, not the access to it in the first place. If you have not charged that crime - the crime of making unauthorized *use* of the data - or if that action is not illegal, your case may be dismissed. In *State v. Lawrence Olson*, 47 Wash. App. 514, 735 P.2d 1362 (1987), the defendant was a police officer. As part of his job he had access to records maintained in the computer system of the state Department of Licensing. He obtained information about University of Washington coeds using that system. This was not related to any investigation. He was charged and convicted of computer trespass under a statute that made criminal the intentional access to a computer system of another without authority. The court, in reversing the conviction, noted that the statute criminalizes the entry into the computer base, not the use of the information obtained. The court also noted that there was no evidence that permission to access the computer was conditioned on the uses made of the data. As a result, although the defendant had made unauthorized use of materials he had obtained from the computer system, he had not committed the crime with which he was charged. The court did note that there could be situations where there are conditions attached to computer access, but there were no such conditions in the case before them.

Computer trespass by otherwise authorized users is difficult to prove. Clear warnings to authorized users about the scope of their authority, both as to areas of the computer they may access, and purpose for which they may access the computer, are essential to any prosecution of a qualified or limited access case.

5. Knowledge/Intent

One common issue in child pornography cases is proving the defendant knew she/he had the offending items on her/his computer. While many news viewers allow you to view a file before you're actually download it to your computer, many others require you to download a file before you can view it. Many news viewers allow you to download multiple files without seeing the contents of any of them. Thus it is quite plausible that someone downloading adult pornography might have what qualifies as child pornography on their computer without their knowledge.

How to you separate the innocent from the guilty? How do you prove knowledge or

intent?

A. Directory Structure

One way is the directory structure. The directory structure can provide clues in at least two different ways.

Directories are files and directory creation date and time is stored with the directory just like file creation date and time is stored with a file. In fact, if the files are copied from one location to another, it is often the directory creation date on the target computer (which shows the date/time the directory was created) that is helpful rather than file creation date/time (since that will usually show the original creation date, not the date it was copied to the target system.)

In addition, the names and organization of the files and directories can show knowledge and intent. In one case in Washington State, *State v. Ronald Rosul*, King County Superior Court No. 96-1-05106-0 SEA, the defendant was accused of possession of child pornography. While an employee of Microsoft the defendant had used Microsoft equipment to burn a CD-ROM containing sexually explicit images, including images of children. His defense was that he had downloaded the material from newsgroups and while he was aware it was sexually explicit material, he was unaware any of the images were of children.

The defendant acknowledged that he had created the directories both on his computer at Microsoft and on the CD-ROM. That directory structure contained a directory called E:\GIFS. Subdirectories included E:\GIFS\BOY and E:\GIFS\YOUNG. This last subdirectory was subdivided as follows:

E:\GIFS\YOUNG\BOY
E:\GIFS\YOUNG\CLEAN
E:\GIFS\YOUNG\YOUNG
E:\GIFS\YOUNG\GIRL.

The E:\GIFS\YOUNG\GIRL subdirectory was subdivided:

E:\GIFS\YOUNG\GIRL\AA
E:\GIFS\YOUNG\GIRL\NUDE
E:\GIFS\YOUNG\GIRL\SEX
E:\GIFS\YOUNG\GIRL\SLYGS

The images in issue were found in the **E:\GIFS\YOUNG\GIRL\SEX** directory. That directory consisted of children under the age of 14 being sexually molested or otherwise sexually exploited. The other directories contained images consistent with the directory name.

The state was able to successfully argue that the defendant was aware of the contents of the images by virtue of the sorting and directory naming he admitted engaging in.

B. File Names

Another was is to examine the file names. If the file names describe their contents (babyfuk.jpg was one file in the Rosul case) there is at least circumstantial evidence that the defendant was aware of the file's contents. This is particularly true if the defendant named the file her/himself.

Often file names, when coupled with an inculpatory directory structure, will be sufficient to show intent or knowledge.

C. Date and Time

Sometimes a file creation date (or the date/time shown on email headers) can be coupled with file names and directory structure to produce an incriminating picture. In the Rosul case there were system logs showing the defendant had accessed a usenet site in the Netherlands several times. The log showed the defendant had downloaded several graphic files in the GIF format. One of the images at the site was found on Rosul's computer, converted to JPG format. The file creation date/time on the JPG file was consistent with the time the log showed Rosul had accessed that same file on the Internet. Producing an expert who testified that such a conversion requires viewing the file, the State was able to demonstrate the defendant's knowledge of the information on his computer.

Conclusion

Searches of computers, whether because of cybercrimes or because the computer contains evidence of other crimes, is becoming common. The law in non-computer search cases provides guidelines that can safely be used in most computer searches. The slowly increasing body of law specific to computer searches will help bridge the gap between non-computer search law and computer search law.

But computer searches are, in some ways, different from non-computer searches. And cases specific to computer searches are still rare. As computer search law evolves there may well be different standards applied to searches of computers.

A practitioner, be they police, prosecutor, defense attorney or judge, should remain alert to the development of the law in this evolving area.